Secure Campus Network Design & Implementation Report

Course: CSCE342301 - Introduction to Cybersecurity (Fall 2025)
Project: Secure Campus Network – Multi-Building Architecture
Team Members: *Ramy Shehata, Omar Ganna, Adham Gohar*
Packet Tracer File: *Project.pkt*

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1. Introduction

This project implements a **secure, multi-campus university network** using Cisco Packet Tracer.
The design includes:

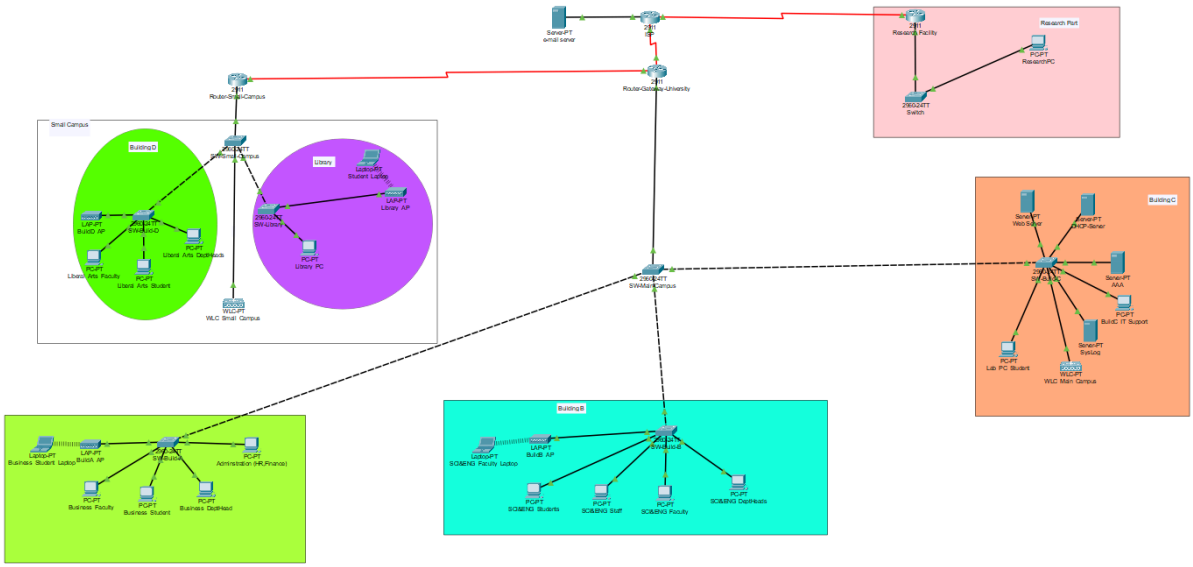
- Main Campus (Buildings A, B, C)
- Small Campus (Building D + Library)
- Remote Research Facility (VPN Site)
- Wireless infrastructure with multiple SSIDs
- DHCP, DNS, Web, AAA, and Syslog servers
- Inter-VLAN routing and ACL-based segmentation
- Secure management VLAN with SSH-only administrator access
- Full security hardening (port security, DHCP snooping, DAI, WPA2, VPN)

This report documents the architecture, configurations, security decisions, risk assessment, and validation screenshots.

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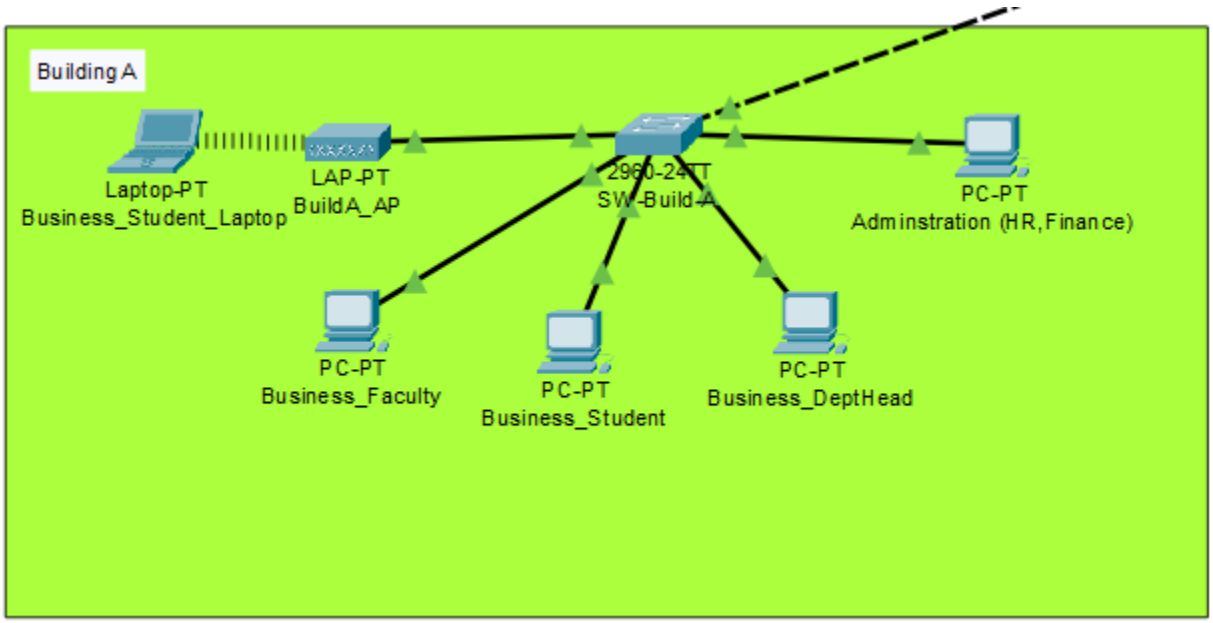
2. Network Architecture

Below is the final logical network topology showing all switches, routers, servers, VLAN segments, APs, and WLCs.

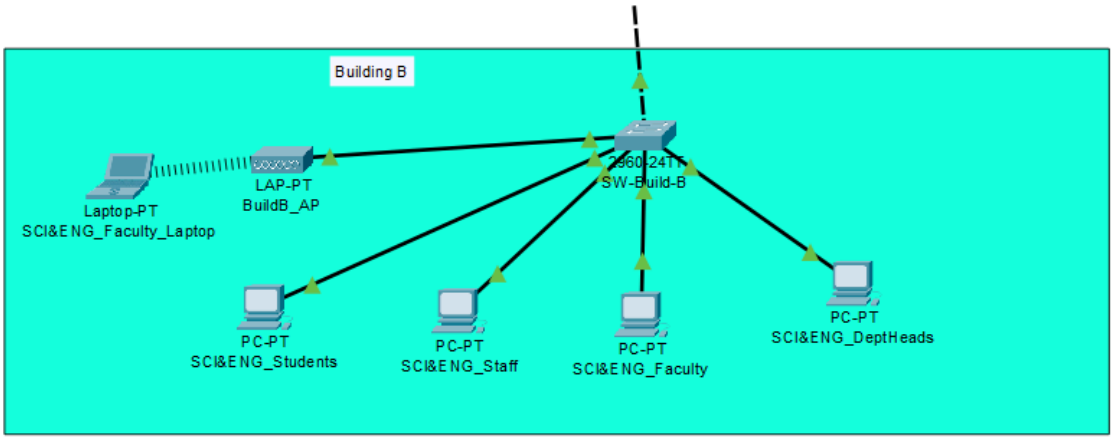


We can go into each building's topology as follows:

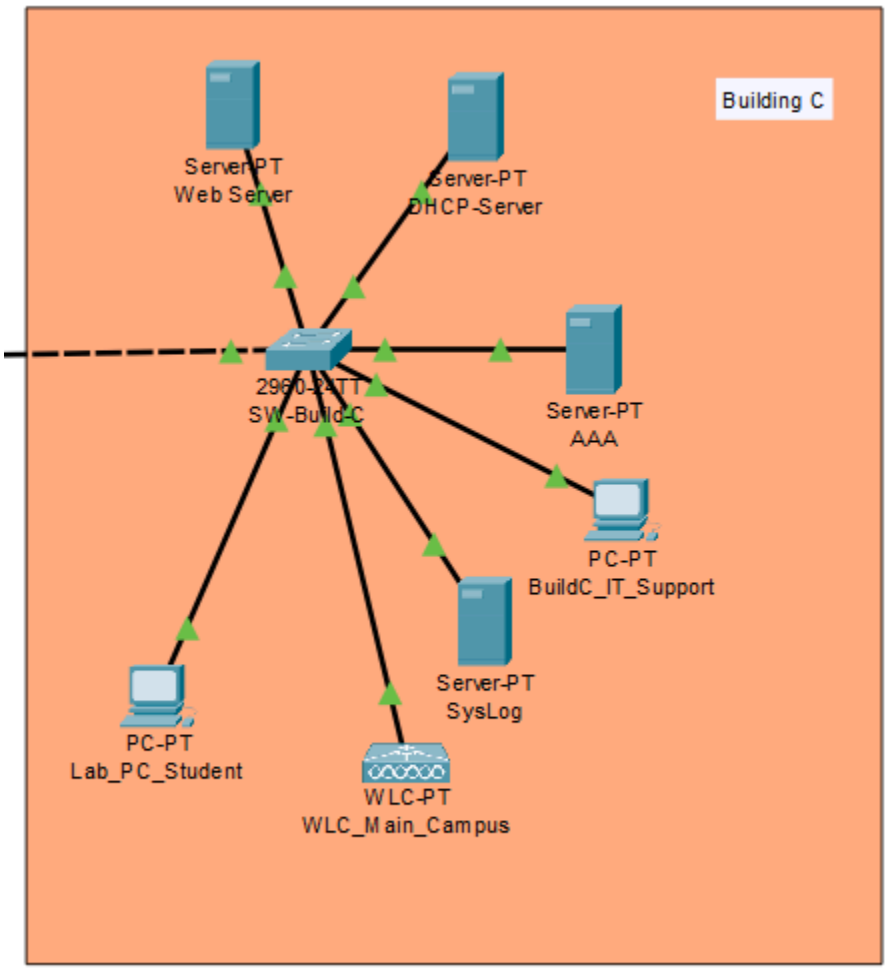
Building A:



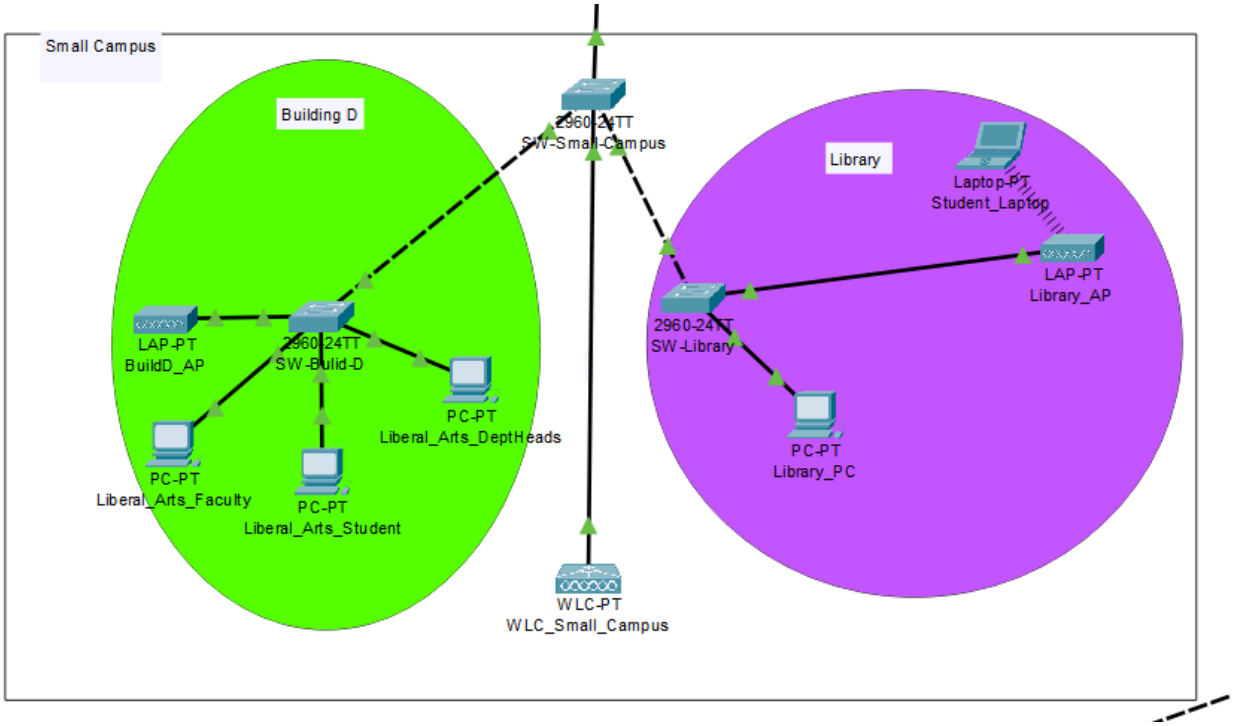
Building B:



Building C:



Small Campus:



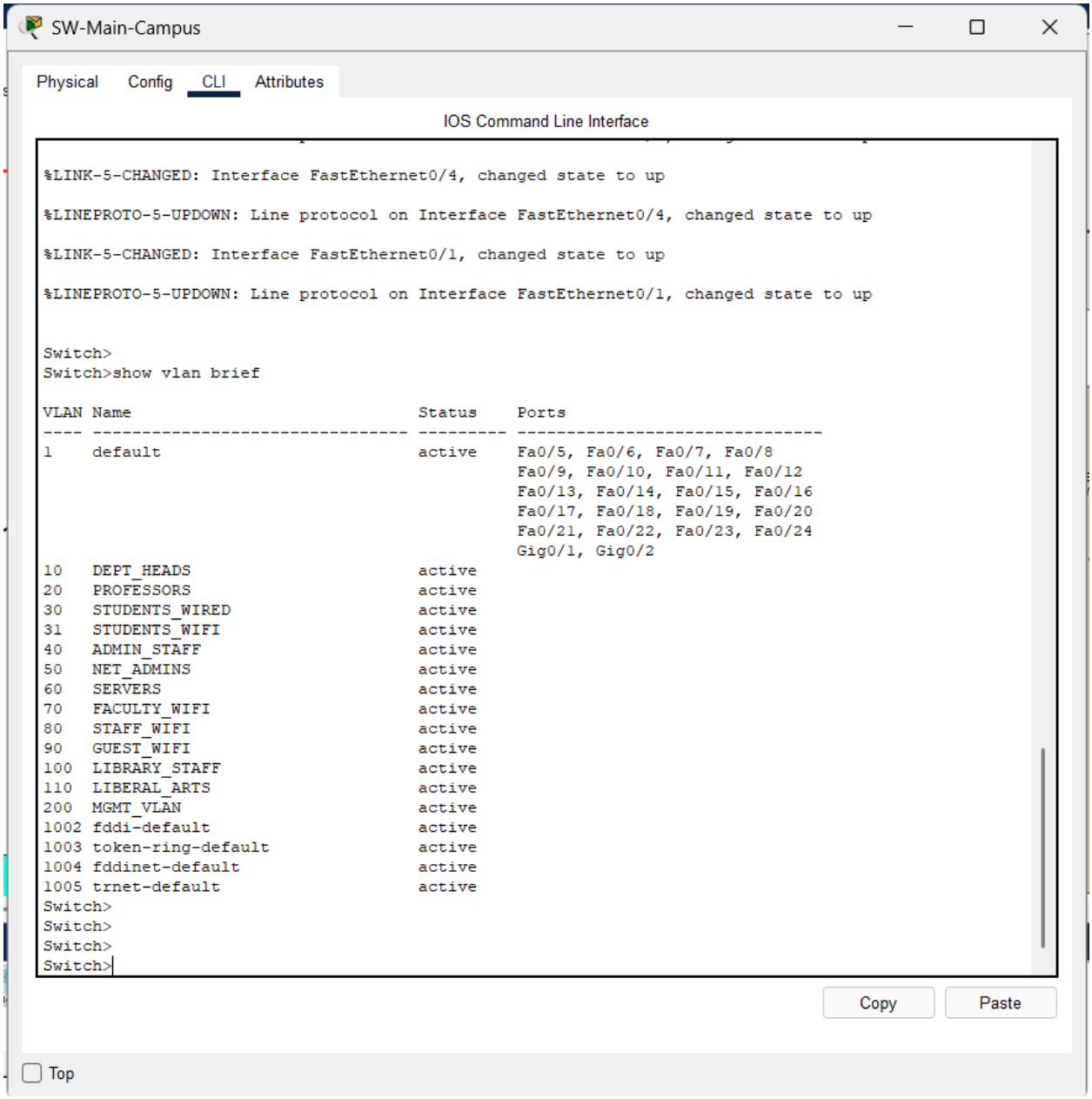
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3. VLAN Table (IDs, Roles, Subnets, SSIDs)

Main Campus:

VLAN Name	VLAN ID	Subnet	Purpose / Role	SSID
Admin Staff	40	**10.20.40.0/24**	HR/Finance/Admin	StaffNet
Network Admin (MGMT)	200	**10.20.200.0/24**	SSH/AAA/SNMP/Syslog	—
Students Wired	30	**10.20.30.0/24**	Student Labs	StudentNet
Students WiFi	31	**10.20.31.0/24**	Student Wireless	StudentNet

Faculty WiFi	70	**10.20.70.0/24**	Faculty Members	FacultyNet
Staff WiFi	80	**10.20.80.0/24**	Administrative WiFi	StaffNet
Guest WiFi	90	**10.20.90.0/24**	Guest Internet	GuestNet
Department Heads	10	**10.20.10.0/24**	Restricted High-Level Access	—
Professors	20	**10.20.20.0/24**	Professors and Faculty	



Small Campus:

VLAN Name	VLAN ID	Subnet	Purpose / Role	SSID
Admin Staff	40	**10.20.45.0/24**	HR/Finance/Admin	StaffNets
Network Admin (MGMT)	200	**10.20.205.0/24**	SSH/AAA/SNMP/Syslog	—

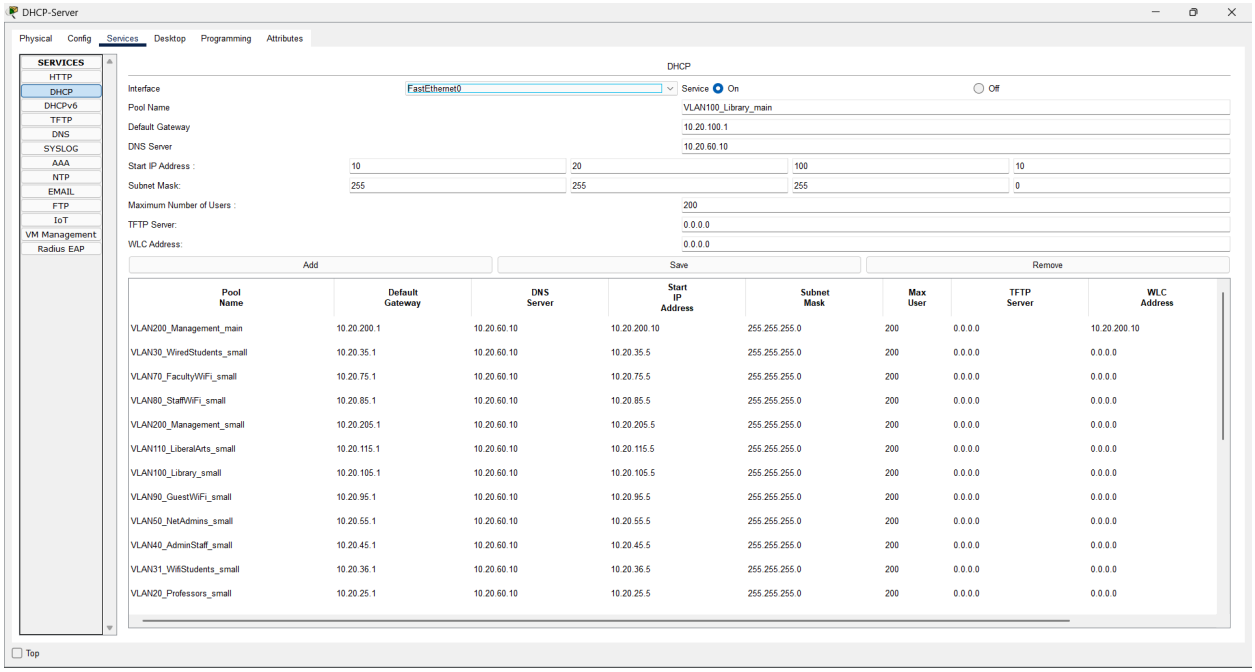
Students Wired	30	**10.20.35.0/24**	Student Labs	StudentNets
Students WiFi	31	**10.20.36.0/24**	Student Wireless	StudentNets
Faculty WiFi	70	**10.20.75.0/24**	Faculty Members	FacultyNets
Staff WiFi	80	**10.20.85.0/24**	Administrative WiFi	StaffNets
Guest WiFi	90	**10.20.95.0/24**	Guest Internet	GuestNets
Department Heads	10	**10.20.15.0/24**	Restricted High-Level Access	—
Professors	20	**10.20.20.0/24**	Professors and Faculty	

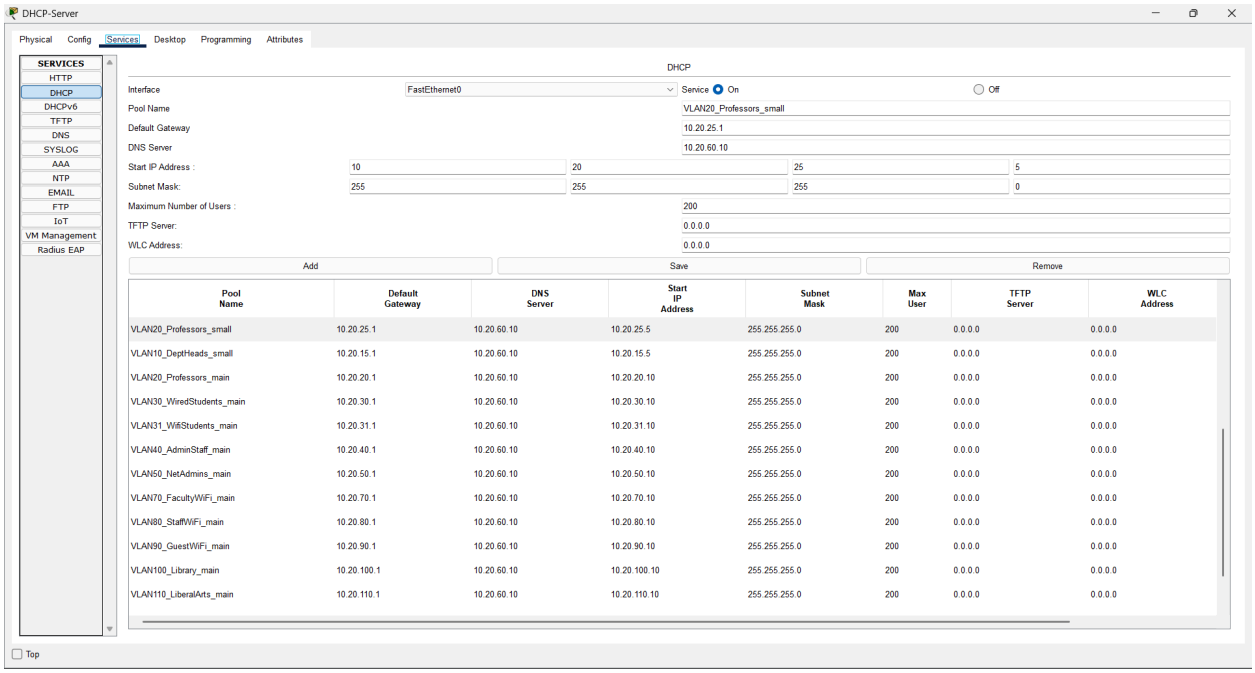
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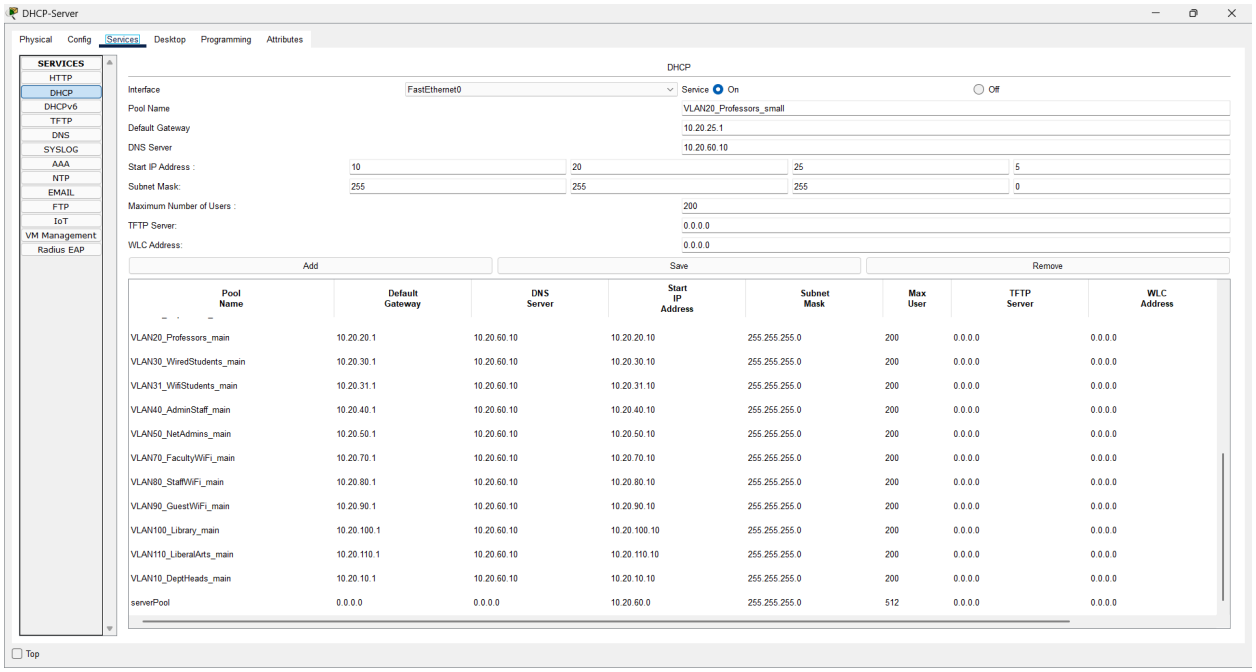
4. IP Addressing Scheme

Summaries of the campus IP plan:

- Base Network: **10.20.0.0/16**
- Point-to-point links: **/30 subnets**
- VLANs each allocated a /24 or /25 depending on size
- DHCP Server in Building C serves all networks via DHCP Relay


DHCP-Server
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
DHCP
Interface
FastEthernet0
Service
On
Off
Pool Name
VLAN100_Library_main
Default Gateway
10.20.100.1
DNS Server
10.20.60.10
Start IP Address :
10
20
100
10
Subnet Mask:
255
255
255
0
Maximum Number of Users :
200
TFTP Server:
0.0.0.0
WLC Address:
0.0.0.0
Add
Save
Remove
Pool Name
Default Gateway
DNS Server
Start IP Address
Subnet Mask
Max User
TFTP Server
WLC Address
VLAN200_Management_main
10.20.200.1
10.20.60.10
10.20.200.10
255.255.255.0
200
0.0.0.0
10.20.200.10
VLAN30_WiredStudents_small
10.20.35.1
10.20.60.10
10.20.35.5
255.255.255.0
200
0.0.0.0
0.0.0.0
VLAN70_FacultyWiFi_small
10.20.75.1
10.20.60.10
10.20.75.5
255.255.255.0
200
0.0.0.0
0.0.0.0
VLAN80_StaffWiFi_small
10.20.85.1
10.20.60.10
10.20.85.5
255.255.255.0
200
0.0.0.0
0.0.0.0
VLAN200_Management_small
10.20.205.1
10.20.60.10
10.20.205.5
255.255.255.0
200
0.0.0.0
0.0.0.0
VLAN110_LiberalArts_small
10.20.115.1
10.20.60.10
10.20.115.5
255.255.255.0
200
0.0.0.0
0.0.0.0
VLAN100_Library_small
10.20.105.1
10.20.60.10
10.20.105.5
255.255.255.0
200
0.0.0.0
0.0.0.0
VLAN90_GuestWiFi_small
10.20.95.1
10.20.60.10
10.20.95.5
255.255.255.0
200
0.0.0.0
0.0.0.0
VLAN50_NetAdmins_small
10.20.55.1
10.20.60.10
10.20.55.5
255.255.255.0
200
0.0.0.0
0.0.0.0
VLAN40_AdminStaff_small
10.20.45.1
10.20.60.10
10.20.45.5
255.255.255.0
200
0.0.0.0
0.0.0.0
VLAN31_WifiStudents_small
10.20.36.1
10.20.60.10
10.20.36.5
255.255.255.0
200
0.0.0.0
0.0.0.0
VLAN20_Professors_small
10.20.25.1
10.20.60.10
10.20.25.5
255.255.255.0
200
0.0.0.0
0.0.0.0
Top


DHCP-Server
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
DHCP
Interface
FastEthernet0
Service
On
Off
Pool Name
VLAN20_Professors_small
Default Gateway
10.20.25.1
DNS Server
10.20.60.10
Start IP Address :
10
20
25
5
Subnet Mask:
255
255
255
0
Maximum Number of Users :
200
TFTP Server:
0.0.0.0
WLC Address:
0.0.0.0
Add
Save
Remove
Pool Name
Default Gateway
DNS Server
Start IP Address
Subnet Mask
Max User
TFTP Server
WLC Address
VLAN20_Professors_small
10.20.25.1
10.20.60.10
10.20.25.5
255.255.255.0
200
0.0.0.0
0.0.0.0
VLAN10_DeptHeads_small
10.20.15.1
10.20.60.10
10.20.15.5
255.255.255.0
200
0.0.0.0
0.0.0.0
VLAN20_Professors_main
10.20.20.1
10.20.60.10
10.20.20.10
255.255.255.0
200
0.0.0.0
0.0.0.0
VLAN30_WiredStudents_main
10.20.30.1
10.20.60.10
10.20.30.10
255.255.255.0
200
0.0.0.0
0.0.0.0
VLAN31_WifiStudents_main
10.20.31.1
10.20.60.10
10.20.31.10
255.255.255.0
200
0.0.0.0
0.0.0.0
VLAN40_AdminStaff_main
10.20.40.1
10.20.60.10
10.20.40.10
255.255.255.0
200
0.0.0.0
0.0.0.0
VLAN50_NetAdmins_main
10.20.50.1
10.20.60.10
10.20.50.10
255.255.255.0
200
0.0.0.0
0.0.0.0
VLAN70_FacultyWiFi_main
10.20.70.1
10.20.60.10
10.20.70.10
255.255.255.0
200
0.0.0.0
0.0.0.0
VLAN80_StaffWiFi_main
10.20.80.1
10.20.60.10
10.20.80.10
255.255.255.0
200
0.0.0.0
0.0.0.0
VLAN90_GuestWiFi_main
10.20.90.1
10.20.60.10
10.20.90.10
255.255.255.0
200
0.0.0.0
0.0.0.0
VLAN100_Library_main
10.20.100.1
10.20.60.10
10.20.100.10
255.255.255.0
200
0.0.0.0
0.0.0.0
VLAN110_LiberalArts_main
10.20.110.1
10.20.60.10
10.20.110.10
255.255.255.0
200
0.0.0.0
0.0.0.0
Top



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5. Device Configuration Screenshots

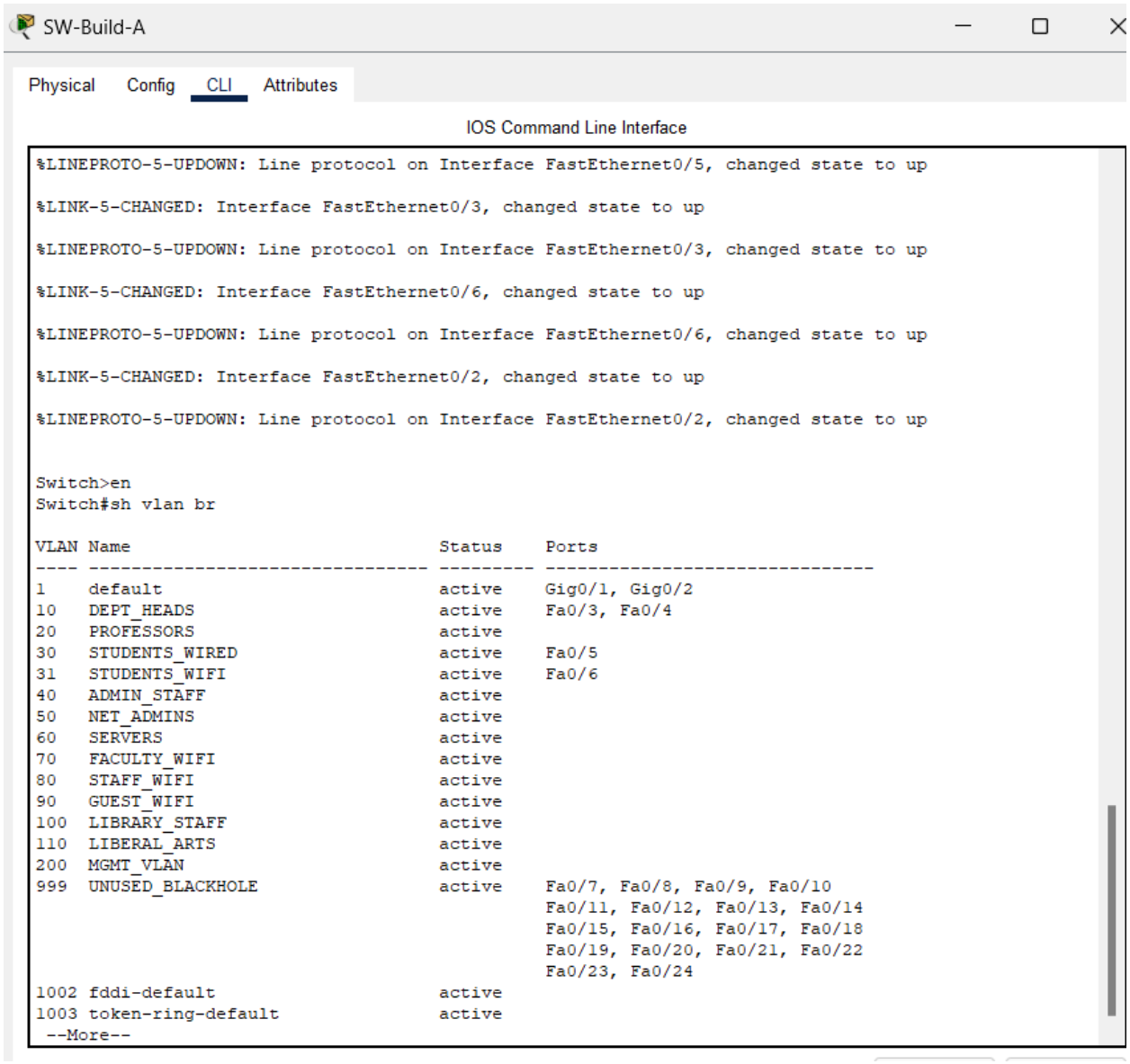
5.1. Switch Configuration (VLANs, Trunks, Port Security)

Commands to Capture

VLAN Creation

```
Switch>en
Switch#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gig0/1, Gig0/2
10   DEPT_HEADS                       active
20   PROFESSORS                       active
30   STUDENTS_WIRED                   active
31   STUDENTS_WIFI                    active
40   ADMIN_STAFF                      active
50   NET_ADMINS                       active
60   SERVERS                          active
70   FACULTY_WIFI                     active
80   STAFF_WIFI                       active
90   GUEST_WIFI                       active
100  LIBRARY_STAFF                    active
110  LIBERAL_ARTS                     active
200  MGMT_VLAN                        active
999  UNUSED_BLACKHOLE                 active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
1002 fddi-default                     active
1003 token-ring-default               active
 --More--
```


SW-Build-A
Physical
Config
CLI
Attributes
IOS Command Line Interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
Switch>en
Switch#sh vlan br
VLAN Name Status Ports
1 default active Gig0/1, Gig0/2
10 DEPT_HEADS active Fa0/3, Fa0/4
20 PROFESSORS active
30 STUDENTS_WIRED active Fa0/5
31 STUDENTS_WIFI active Fa0/6
40 ADMIN_STAFF active
50 NET_ADMINS active
60 SERVERS active
70 FACULTY_WIFI active
80 STAFF_WIFI active
90 GUEST_WIFI active
100 LIBRARY_STAFF active
110 LIBERAL_ARTS active
200 MGMT_VLAN active
999 UNUSED_BLACKHOLE active Fa0/7, Fa0/8, Fa0/9, Fa0/10
Fa0/11, Fa0/12, Fa0/13, Fa0/14
Fa0/15, Fa0/16, Fa0/17, Fa0/18
Fa0/19, Fa0/20, Fa0/21, Fa0/22
Fa0/23, Fa0/24
1002 fddi-default active
1003 token-ring-default active
--More--

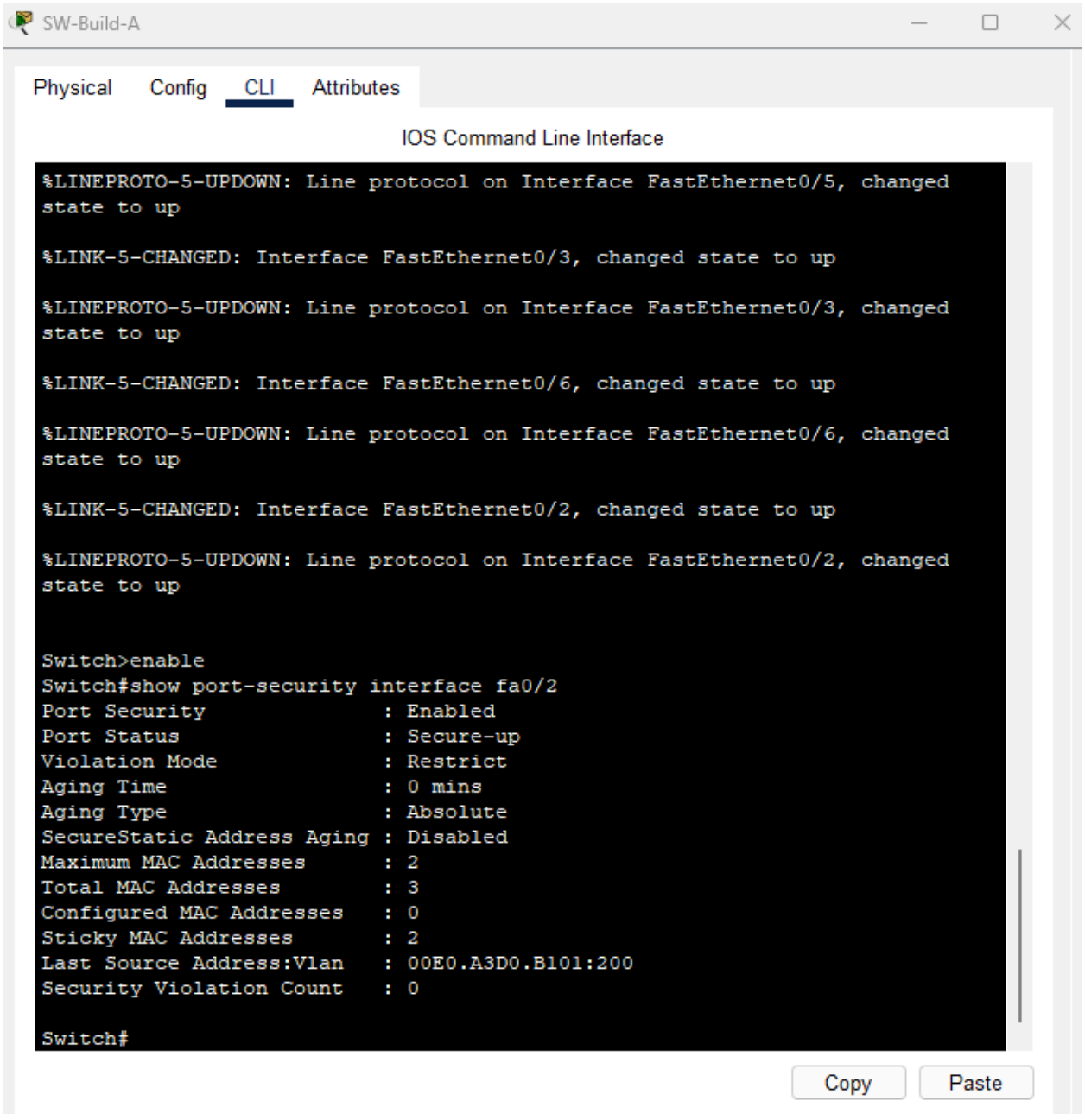
```
Switch#sh vlan br

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Fa0/8, Fa0/9, Gig0/1, Gig0/2
10   DEPT_HEADS                        active    Fa0/6
20   PROFESSORS                        active    Fa0/5
30   STUDENTS_WIRED                    active    Fa0/3
31   STUDENTS_WIFI                     active
40   ADMIN_STAFF                       active    Fa0/4
50   NET_ADMINS                        active
60   SERVERS                           active
70   FACULTY_WIFI                      active
80   STAFF_WIFI                        active
90   GUEST_WIFI                        active
100  LIBRARY_STAFF                     active
110  LIBERAL_ARTS                      active
200  MGMT_VLAN                         active    Fa0/7
999  UNUSED                            active    Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                 Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                 Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                 Fa0/22, Fa0/23, Fa0/24
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
 --More--
```

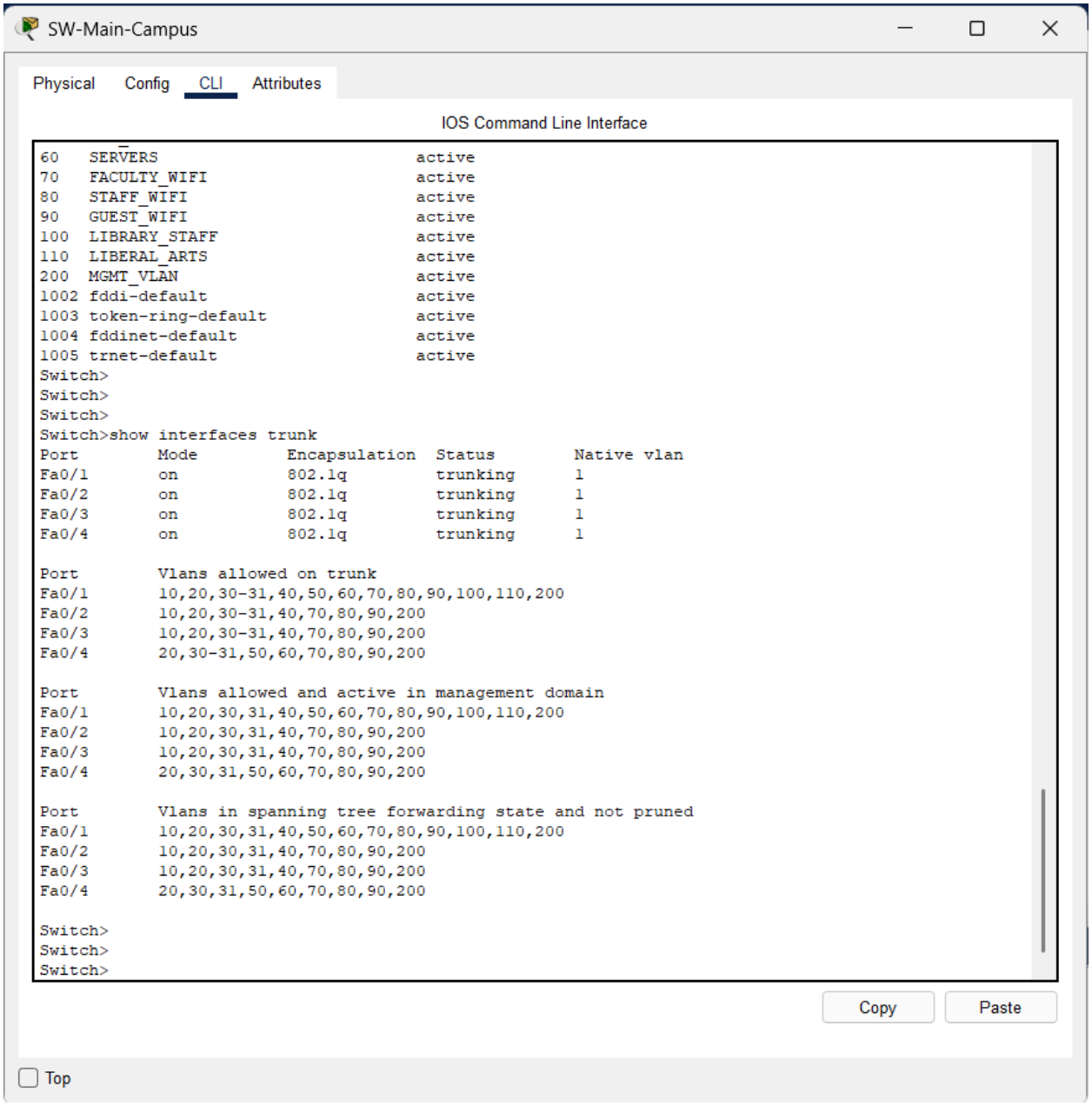
```
Switch#sh vlan br

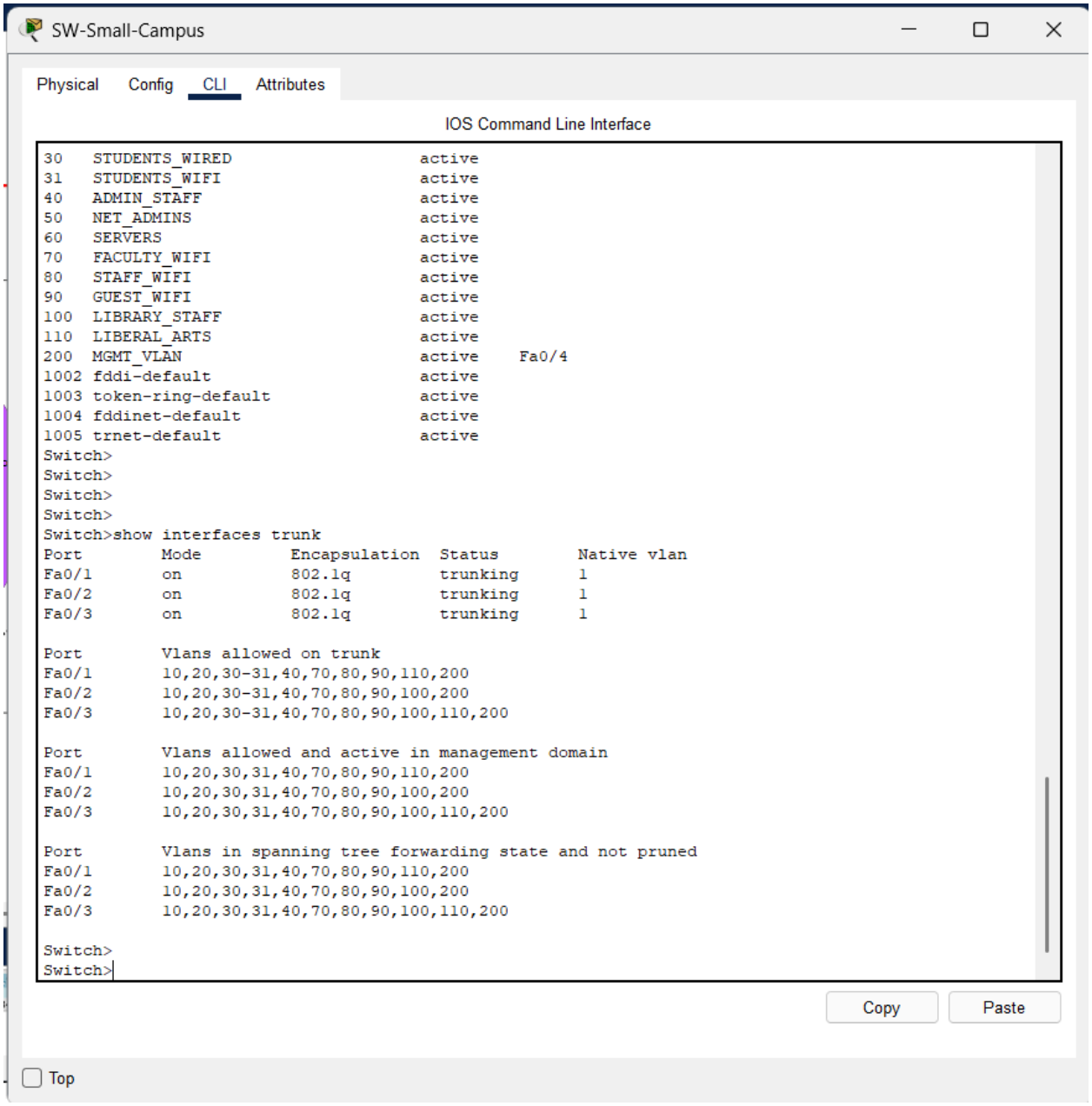
VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Gig0/1, Gig0/2
10   DEPT_HEADS                        active
20   PROFESSORS                        active
30   STUDENTS_WIRED                    active    Fa0/8
31   STUDENTS_WIFI                     active
40   ADMIN_STAFF                       active
50   NET_ADMINS                        active    Fa0/7
60   SERVERS                           active    Fa0/3, Fa0/4, Fa0/5
70   FACULTY_WIFI                      active
80   STAFF_WIFI                        active
90   GUEST_WIFI                        active
100  LIBRARY_STAFF                     active
110  LIBERAL_ARTS                      active
200  MGMT_VLAN                         active
999  VLAN0999                          active    Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                 Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                 Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                 Fa0/21, Fa0/22, Fa0/23, Fa0/24
1002 fddi-default                      active
1003 token-ring-default                active
 --More--
```

Port Security

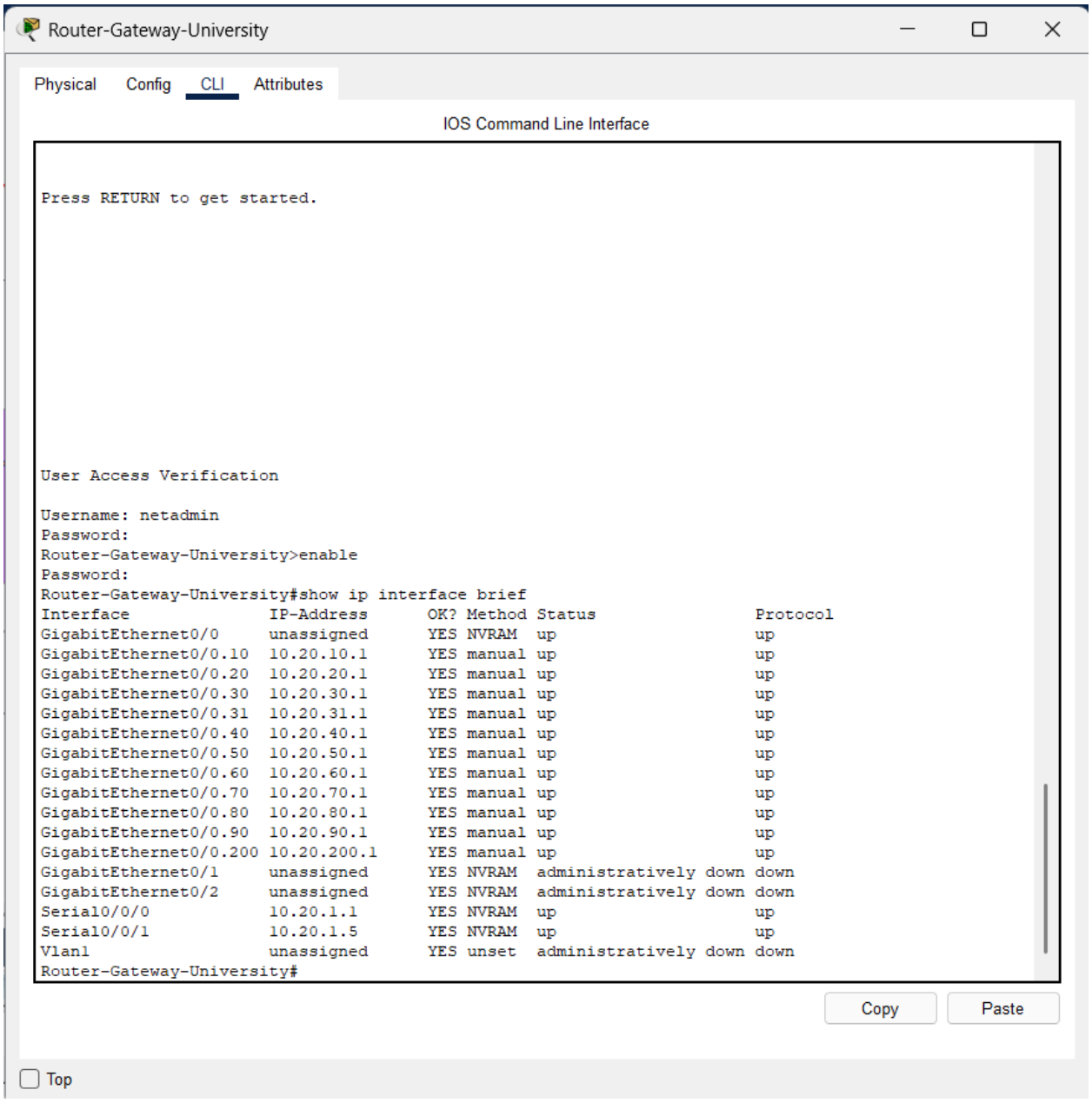

SW-Build-A
Physical
Config
CLI
Attributes
IOS Command Line Interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
Switch>enable
Switch#show port-security interface fa0/2
Port Security : Enabled
Port Status : Secure-up
Violation Mode : Restrict
Aging Time : 0 mins
Aging Type : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses : 2
Total MAC Addresses : 3
Configured MAC Addresses : 0
Sticky MAC Addresses : 2
Last Source Address:Vlan : 00E0.A3D0.B101:200
Security Violation Count : 0
Switch#
Copy
Paste

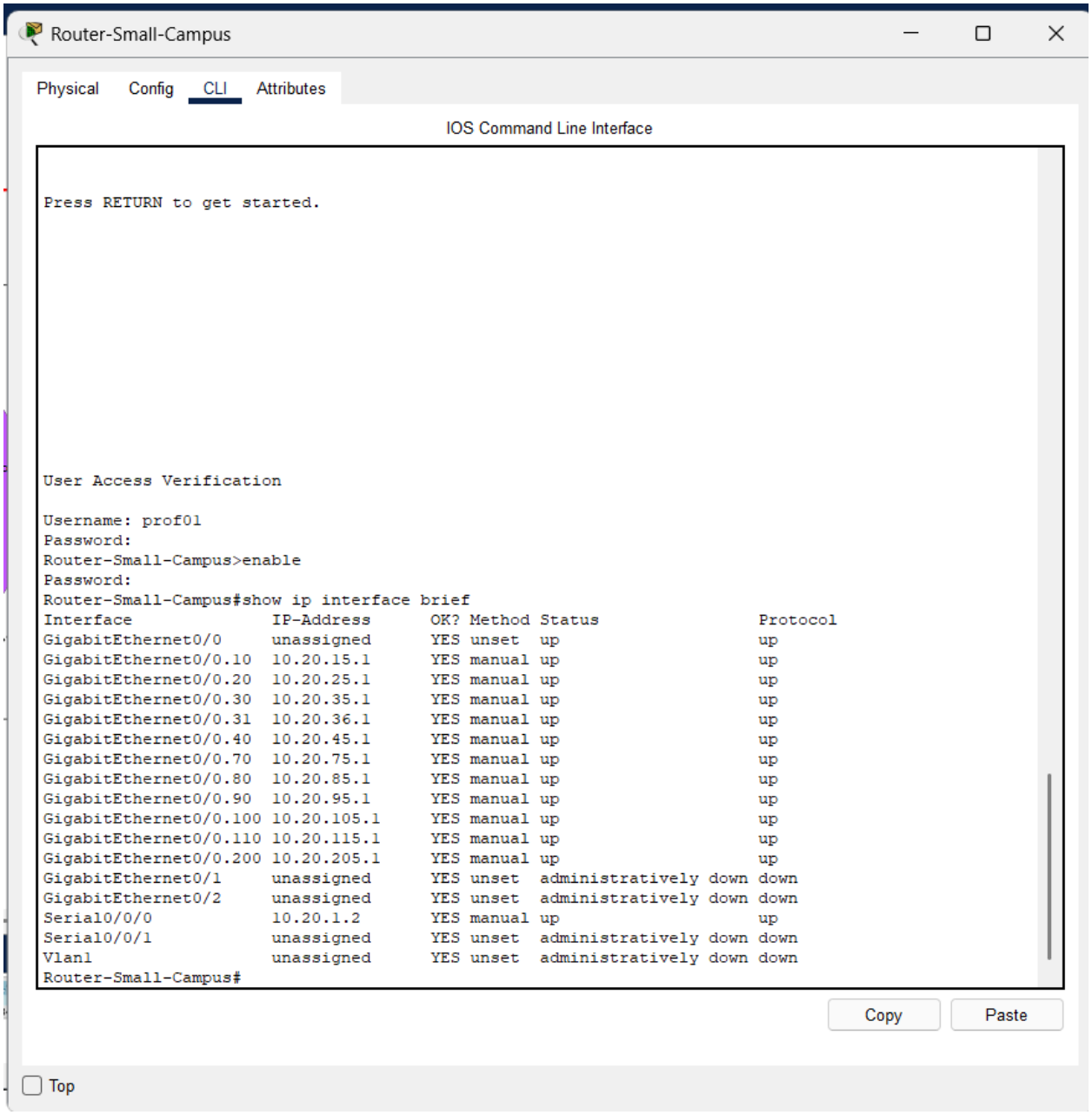
Trunk Links


SW-Main-Campus
Physical
Config
CLI
Attributes
IOS Command Line Interface
60 SERVERS active
70 FACULTY_WIFI active
80 STAFF_WIFI active
90 GUEST_WIFI active
100 LIBRARY_STAFF active
110 LIBERAL_ARTS active
200 MGMT_VLAN active
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
Switch>
Switch>
Switch>
Switch>show interfaces trunk
Port Mode Encapsulation Status Native vlan
Fa0/1 on 802.1q trunking 1
Fa0/2 on 802.1q trunking 1
Fa0/3 on 802.1q trunking 1
Fa0/4 on 802.1q trunking 1
Port Vlans allowed on trunk
Fa0/1 10,20,30-31,40,50,60,70,80,90,100,110,200
Fa0/2 10,20,30-31,40,70,80,90,200
Fa0/3 10,20,30-31,40,70,80,90,200
Fa0/4 20,30-31,50,60,70,80,90,200
Port Vlans allowed and active in management domain
Fa0/1 10,20,30,31,40,50,60,70,80,90,100,110,200
Fa0/2 10,20,30,31,40,70,80,90,200
Fa0/3 10,20,30,31,40,70,80,90,200
Fa0/4 20,30,31,50,60,70,80,90,200
Port Vlans in spanning tree forwarding state and not pruned
Fa0/1 10,20,30,31,40,50,60,70,80,90,100,110,200
Fa0/2 10,20,30,31,40,70,80,90,200
Fa0/3 10,20,30,31,40,70,80,90,200
Fa0/4 20,30,31,50,60,70,80,90,200
Switch>
Switch>
Switch>
Copy
Paste
Top



5.2. Inter-VLAN Routing (Router-on-a-Stick)


Router-Gateway-University
Physical
Config
CLI
Attributes
IOS Command Line Interface
Press RETURN to get started.
User Access Verification
Username: netadmin
Password:
Router-Gateway-University>enable
Password:
Router-Gateway-University#show ip interface brief
Interface   IP-Address   OK? Method Status   Protocol
GigabitEthernet0/0   unassigned   YES NVRAM up   up
GigabitEthernet0/0.10 10.20.10.1   YES manual up   up
GigabitEthernet0/0.20 10.20.20.1   YES manual up   up
GigabitEthernet0/0.30 10.20.30.1   YES manual up   up
GigabitEthernet0/0.31 10.20.31.1   YES manual up   up
GigabitEthernet0/0.40 10.20.40.1   YES manual up   up
GigabitEthernet0/0.50 10.20.50.1   YES manual up   up
GigabitEthernet0/0.60 10.20.60.1   YES manual up   up
GigabitEthernet0/0.70 10.20.70.1   YES manual up   up
GigabitEthernet0/0.80 10.20.80.1   YES manual up   up
GigabitEthernet0/0.90 10.20.90.1   YES manual up   up
GigabitEthernet0/0.200 10.20.200.1   YES manual up   up
GigabitEthernet0/1   unassigned   YES NVRAM administratively down down
GigabitEthernet0/2   unassigned   YES NVRAM administratively down down
Serial0/0/0   10.20.1.1   YES NVRAM up   up
Serial0/0/1   10.20.1.5   YES NVRAM up   up
Vlan1   unassigned   YES unset administratively down down
Router-Gateway-University#
Copy
Paste
Top


Router-Small-Campus
Physical
Config
CLI
Attributes
IOS Command Line Interface
Press RETURN to get started.
User Access Verification
Username: prof01
Password:
Router-Small-Campus>enable
Password:
Router-Small-Campus#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 unassigned YES unset up up
GigabitEthernet0/0.10 10.20.15.1 YES manual up up
GigabitEthernet0/0.20 10.20.25.1 YES manual up up
GigabitEthernet0/0.30 10.20.35.1 YES manual up up
GigabitEthernet0/0.31 10.20.36.1 YES manual up up
GigabitEthernet0/0.40 10.20.45.1 YES manual up up
GigabitEthernet0/0.70 10.20.75.1 YES manual up up
GigabitEthernet0/0.80 10.20.85.1 YES manual up up
GigabitEthernet0/0.90 10.20.95.1 YES manual up up
GigabitEthernet0/0.100 10.20.105.1 YES manual up up
GigabitEthernet0/0.110 10.20.115.1 YES manual up up
GigabitEthernet0/0.200 10.20.205.1 YES manual up up
GigabitEthernet0/1 unassigned YES unset administratively down down
GigabitEthernet0/2 unassigned YES unset administratively down down
Serial0/0/0 10.20.1.2 YES manual up up
Serial0/0/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router-Small-Campus#
Copy
Paste
Top

Router-Gateway-University

Physical Config CLI Attributes

IOS Command Line Interface

```
Router-Gateway-University#show run | section interface GigabitEthernet0/0
interface GigabitEthernet0/0
 no ip address
 duplex auto
 speed auto
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 10.20.10.1 255.255.255.0
 ip helper-address 10.20.60.10
 ip access-group DEPTHEADS_OUT in
interface GigabitEthernet0/0.20
 description VLAN 20 - Professors
 encapsulation dot1Q 20
 ip address 10.20.20.1 255.255.255.0
 ip helper-address 10.20.60.10
 ip access-group PROFS_OUT in
interface GigabitEthernet0/0.30
 description Small Campus - Students VLAN 30
 encapsulation dot1Q 30
 ip address 10.20.30.1 255.255.255.0
 ip helper-address 10.20.60.10
 ip access-group STUDENTS_OUT in
interface GigabitEthernet0/0.31
 encapsulation dot1Q 31
 ip address 10.20.31.1 255.255.255.0
 ip helper-address 10.20.60.10
 ip access-group STUDENTS_OUT in
interface GigabitEthernet0/0.40
 description VLAN 40 - Admin Staff
 encapsulation dot1Q 40
 ip address 10.20.40.1 255.255.255.0
 ip helper-address 10.20.60.10
 ip access-group ADMIN_OUT in
interface GigabitEthernet0/0.50
 encapsulation dot1Q 50
 ip address 10.20.50.1 255.255.255.0
 ip helper-address 10.20.60.10
interface GigabitEthernet0/0.60
 encapsulation dot1Q 60
 ip address 10.20.60.1 255.255.255.0
 ip helper-address 10.20.60.10
interface GigabitEthernet0/0.70
 encapsulation dot1Q 70
 ip address 10.20.70.1 255.255.255.0
 ip helper-address 10.20.60.10
 ip access-group PROFS_OUT in
interface GigabitEthernet0/0.80
 encapsulation dot1Q 80
 ip address 10.20.80.1 255.255.255.0
 ip helper-address 10.20.60.10
 ip access-group ADMIN_OUT in
```

Top

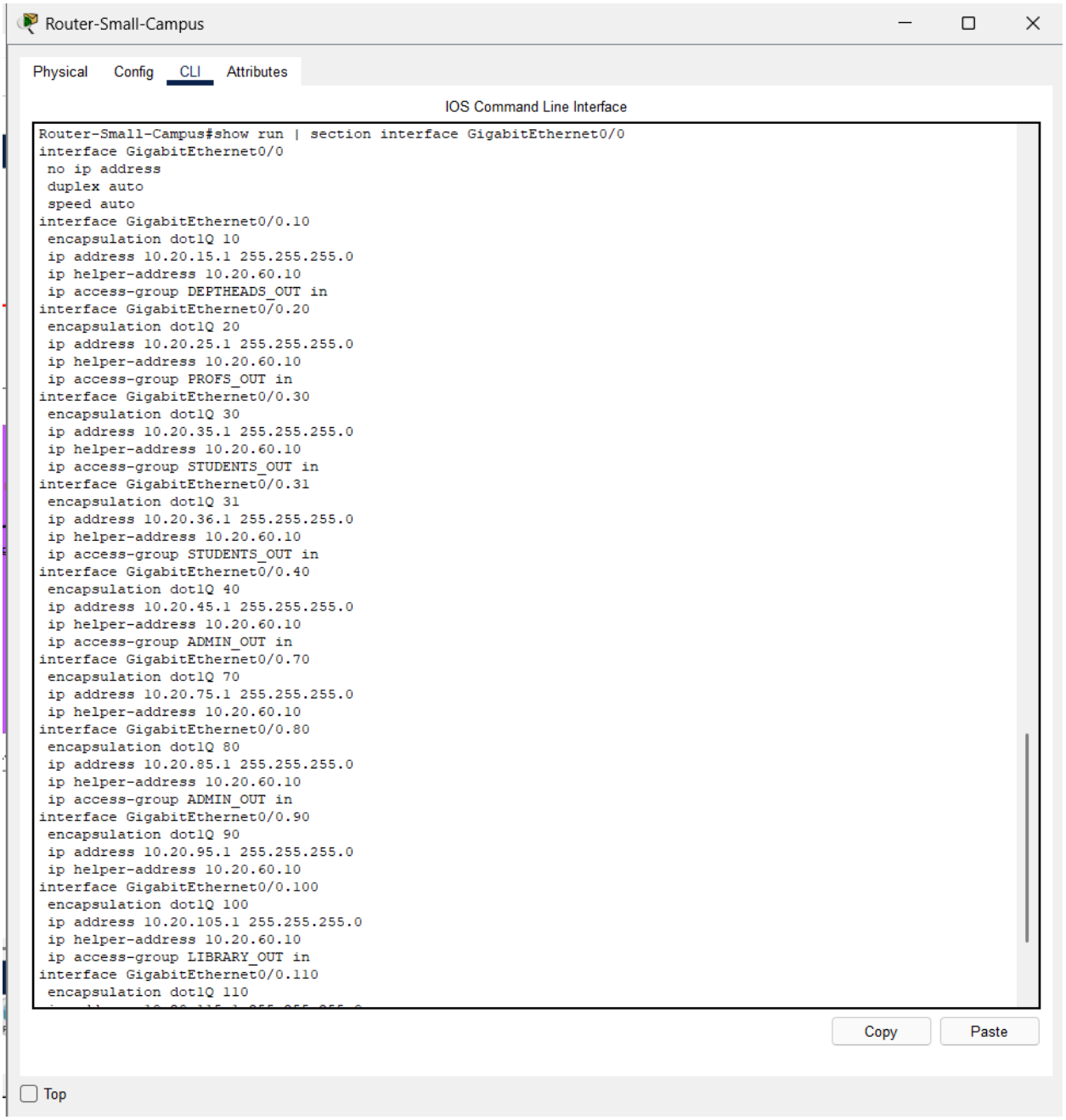
Router-Gateway-University

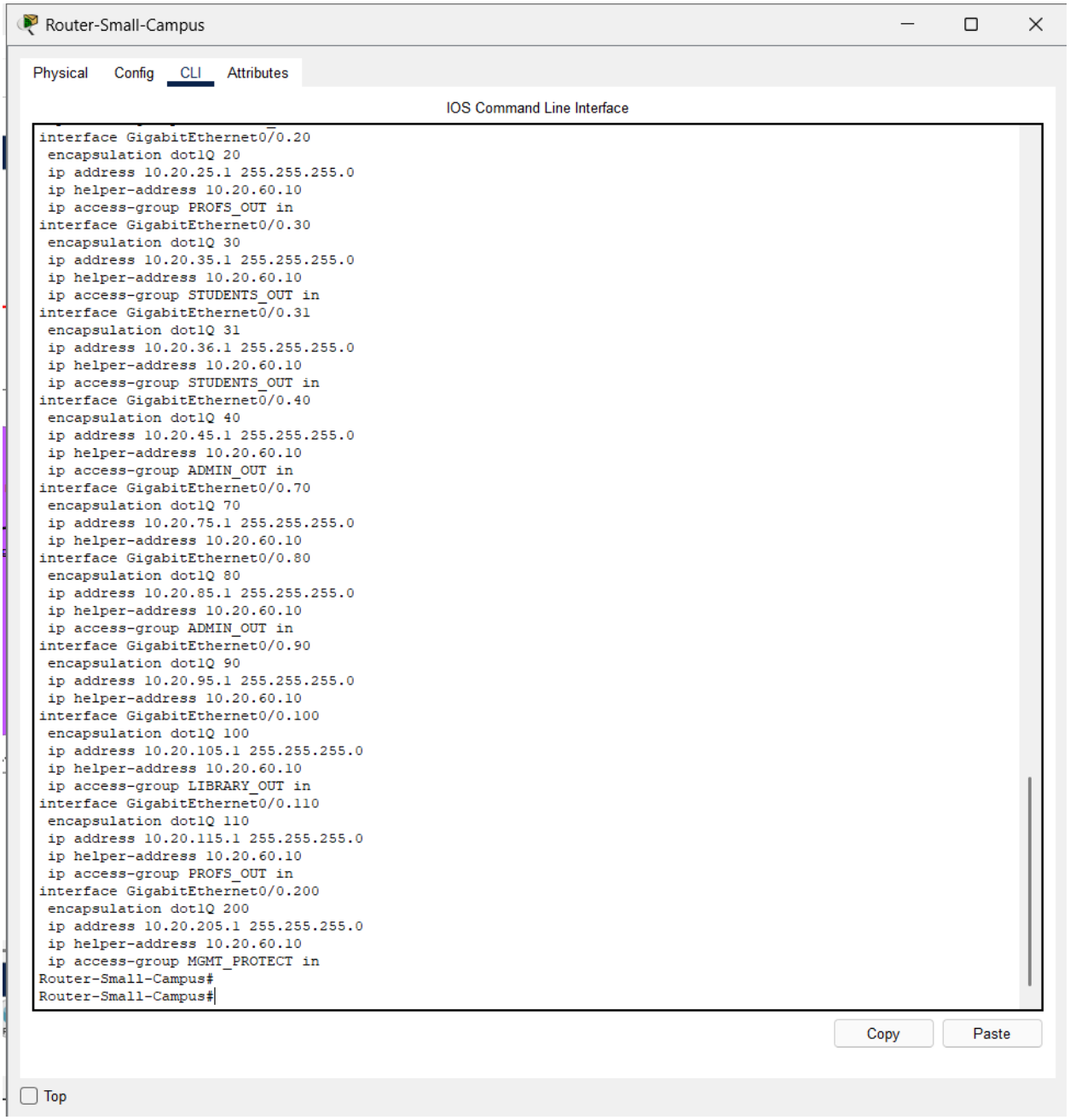
Physical Config CLI Attributes

IOS Com

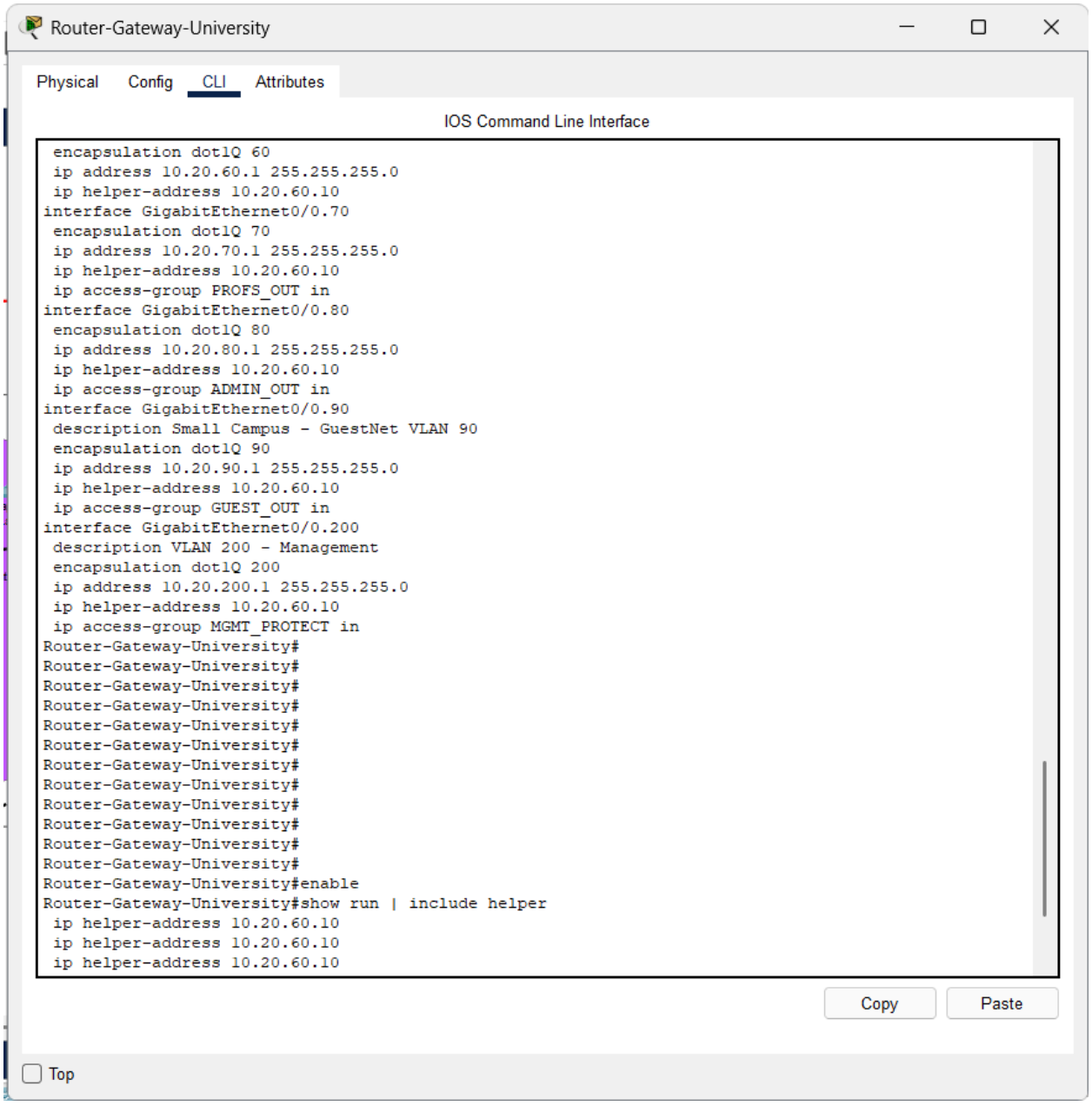
```
 ip address 10.20.31.1 255.255.255.0
 ip helper-address 10.20.60.10
 ip access-group STUDENTS_OUT in
interface GigabitEthernet0/0.40
 description VLAN 40 - Admin Staff
 encapsulation dot1Q 40
 ip address 10.20.40.1 255.255.255.0
 ip helper-address 10.20.60.10
 ip access-group ADMIN_OUT in
interface GigabitEthernet0/0.50
 encapsulation dot1Q 50
 ip address 10.20.50.1 255.255.255.0
 ip helper-address 10.20.60.10
interface GigabitEthernet0/0.60
 encapsulation dot1Q 60
 ip address 10.20.60.1 255.255.255.0
 ip helper-address 10.20.60.10
interface GigabitEthernet0/0.70
 encapsulation dot1Q 70
 ip address 10.20.70.1 255.255.255.0
 ip helper-address 10.20.60.10
 ip access-group PROFS_OUT in
interface GigabitEthernet0/0.80
 encapsulation dot1Q 80
 ip address 10.20.80.1 255.255.255.0
 ip helper-address 10.20.60.10
 ip access-group ADMIN_OUT in
interface GigabitEthernet0/0.90
 description Small Campus - GuestNet VLAN 90
 encapsulation dot1Q 90
 ip address 10.20.90.1 255.255.255.0
 ip helper-address 10.20.60.10
 ip access-group GUEST_OUT in
interface GigabitEthernet0/0.200
 description VLAN 200 - Management
 encapsulation dot1Q 200
 ip address 10.20.200.1 255.255.255.0
 ip helper-address 10.20.60.10
 ip access-group MGMT_PROTECT in
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#
```

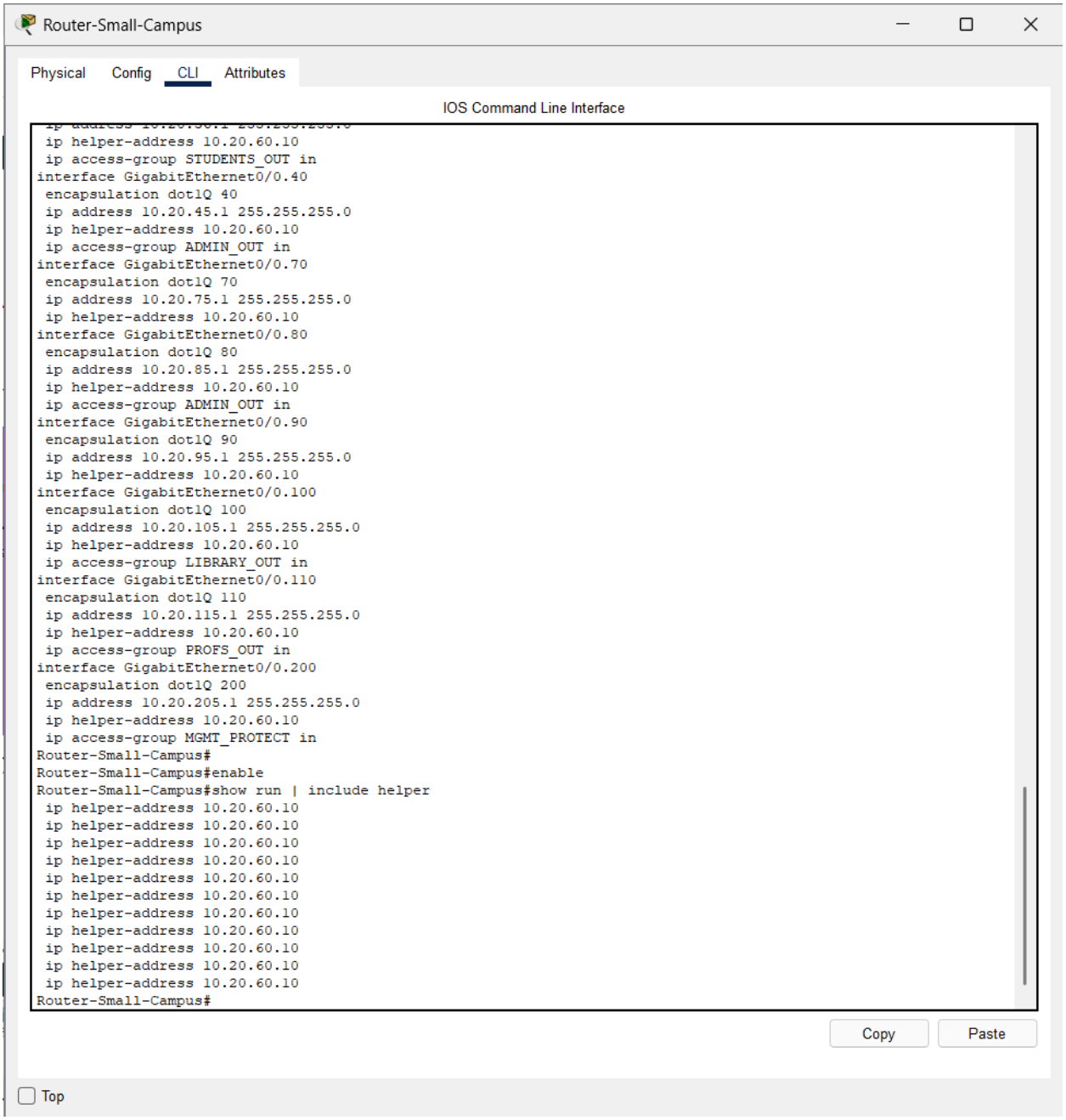
Top

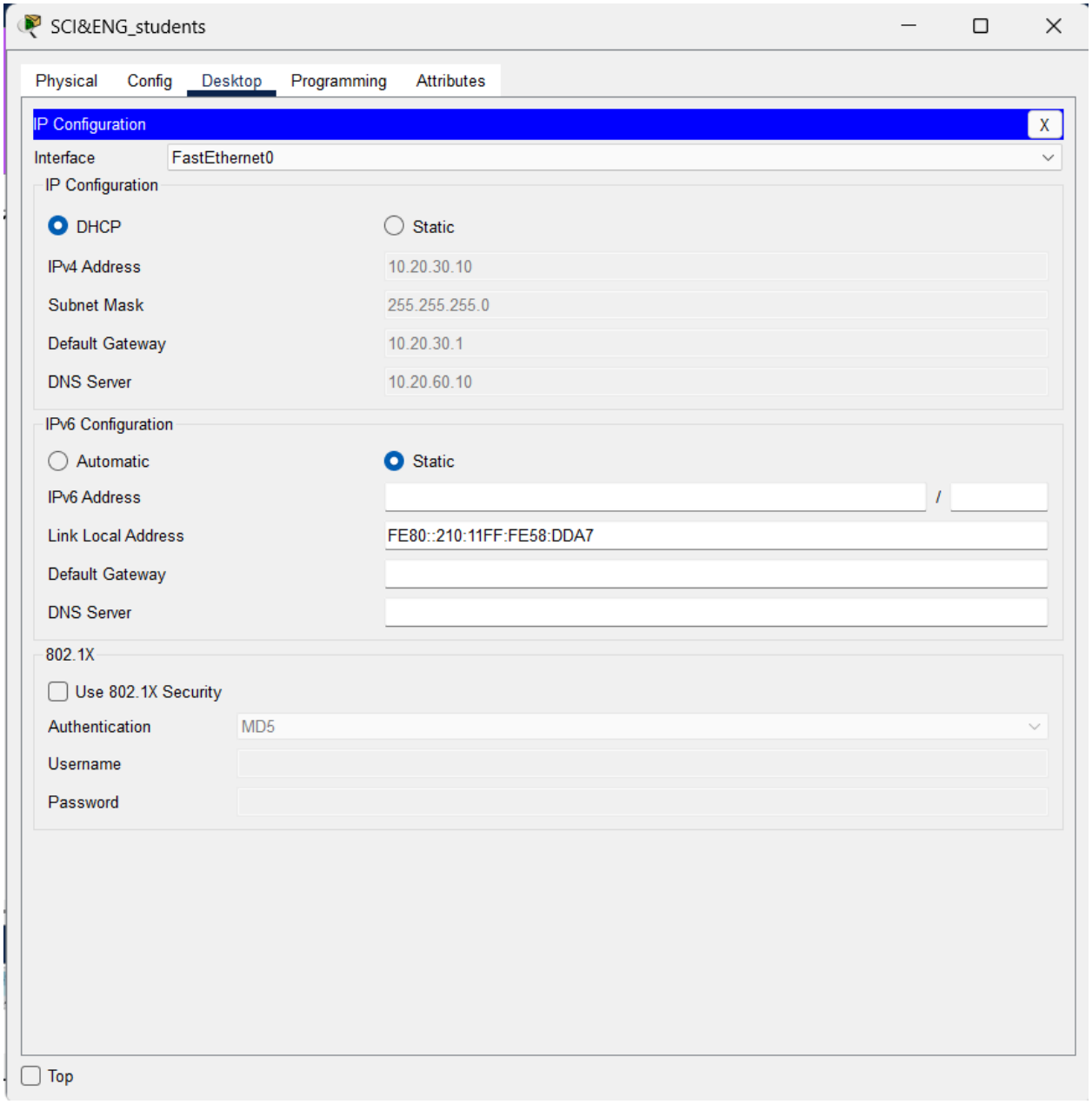

Router-Small-Campus
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router-Small-Campus#show run | section interface GigabitEthernet0/0
interface GigabitEthernet0/0
 no ip address
 duplex auto
 speed auto
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 10.20.15.1 255.255.255.0
 ip helper-address 10.20.60.10
 ip access-group DEPTHEADS_OUT in
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 10.20.25.1 255.255.255.0
 ip helper-address 10.20.60.10
 ip access-group PROFS_OUT in
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 10.20.35.1 255.255.255.0
 ip helper-address 10.20.60.10
 ip access-group STUDENTS_OUT in
interface GigabitEthernet0/0.31
 encapsulation dot1Q 31
 ip address 10.20.36.1 255.255.255.0
 ip helper-address 10.20.60.10
 ip access-group STUDENTS_OUT in
interface GigabitEthernet0/0.40
 encapsulation dot1Q 40
 ip address 10.20.45.1 255.255.255.0
 ip helper-address 10.20.60.10
 ip access-group ADMIN_OUT in
interface GigabitEthernet0/0.70
 encapsulation dot1Q 70
 ip address 10.20.75.1 255.255.255.0
 ip helper-address 10.20.60.10
interface GigabitEthernet0/0.80
 encapsulation dot1Q 80
 ip address 10.20.85.1 255.255.255.0
 ip helper-address 10.20.60.10
 ip access-group ADMIN_OUT in
interface GigabitEthernet0/0.90
 encapsulation dot1Q 90
 ip address 10.20.95.1 255.255.255.0
 ip helper-address 10.20.60.10
interface GigabitEthernet0/0.100
 encapsulation dot1Q 100
 ip address 10.20.105.1 255.255.255.0
 ip helper-address 10.20.60.10
 ip access-group LIBRARY_OUT in
interface GigabitEthernet0/0.110
 encapsulation dot1Q 110
Copy
Paste
Top



5.3. DHCP Relay (ip helper-address)


Router-Gateway-University
Physical
Config
CLI
Attributes
IOS Command Line Interface
encapsulation dot1Q 60
ip address 10.20.60.1 255.255.255.0
ip helper-address 10.20.60.10
interface GigabitEthernet0/0.70
encapsulation dot1Q 70
ip address 10.20.70.1 255.255.255.0
ip helper-address 10.20.60.10
ip access-group PROFS_OUT in
interface GigabitEthernet0/0.80
encapsulation dot1Q 80
ip address 10.20.80.1 255.255.255.0
ip helper-address 10.20.60.10
ip access-group ADMIN_OUT in
interface GigabitEthernet0/0.90
description Small Campus - GuestNet VLAN 90
encapsulation dot1Q 90
ip address 10.20.90.1 255.255.255.0
ip helper-address 10.20.60.10
ip access-group GUEST_OUT in
interface GigabitEthernet0/0.200
description VLAN 200 - Management
encapsulation dot1Q 200
ip address 10.20.200.1 255.255.255.0
ip helper-address 10.20.60.10
ip access-group MGMT_PROTECT in
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#enable
Router-Gateway-University#show run | include helper
ip helper-address 10.20.60.10
ip helper-address 10.20.60.10
ip helper-address 10.20.60.10
Copy
Paste
Top

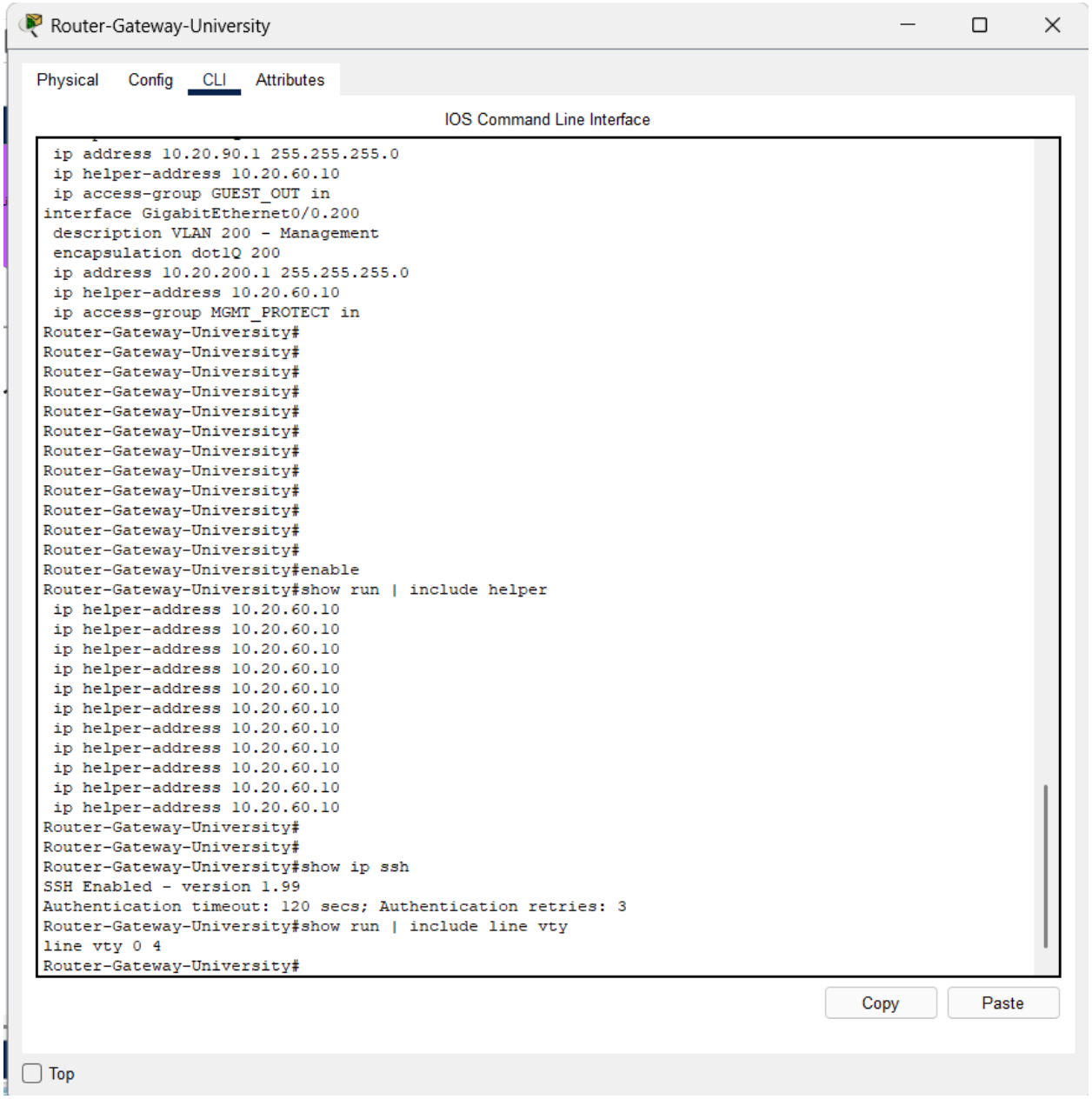

Router-Small-Campus
Physical
Config
CLI
Attributes
IOS Command Line Interface
ip helper-address 10.20.60.10
ip access-group STUDENTS_OUT in
interface GigabitEthernet0/0.40
encapsulation dot1Q 40
ip address 10.20.45.1 255.255.255.0
ip helper-address 10.20.60.10
ip access-group ADMIN_OUT in
interface GigabitEthernet0/0.70
encapsulation dot1Q 70
ip address 10.20.75.1 255.255.255.0
ip helper-address 10.20.60.10
interface GigabitEthernet0/0.80
encapsulation dot1Q 80
ip address 10.20.85.1 255.255.255.0
ip helper-address 10.20.60.10
ip access-group ADMIN_OUT in
interface GigabitEthernet0/0.90
encapsulation dot1Q 90
ip address 10.20.95.1 255.255.255.0
ip helper-address 10.20.60.10
interface GigabitEthernet0/0.100
encapsulation dot1Q 100
ip address 10.20.105.1 255.255.255.0
ip helper-address 10.20.60.10
ip access-group LIBRARY_OUT in
interface GigabitEthernet0/0.110
encapsulation dot1Q 110
ip address 10.20.115.1 255.255.255.0
ip helper-address 10.20.60.10
ip access-group PROFS_OUT in
interface GigabitEthernet0/0.200
encapsulation dot1Q 200
ip address 10.20.205.1 255.255.255.0
ip helper-address 10.20.60.10
ip access-group MGMT_PROTECT in
Router-Small-Campus#
Router-Small-Campus#enable
Router-Small-Campus#show run | include helper
ip helper-address 10.20.60.10
ip helper-address 10.20.60.10
ip helper-address 10.20.60.10
ip helper-address 10.20.60.10
ip helper-address 10.20.60.10
ip helper-address 10.20.60.10
ip helper-address 10.20.60.10
ip helper-address 10.20.60.10
ip helper-address 10.20.60.10
ip helper-address 10.20.60.10
ip helper-address 10.20.60.10
ip helper-address 10.20.60.10
Router-Small-Campus#
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Top

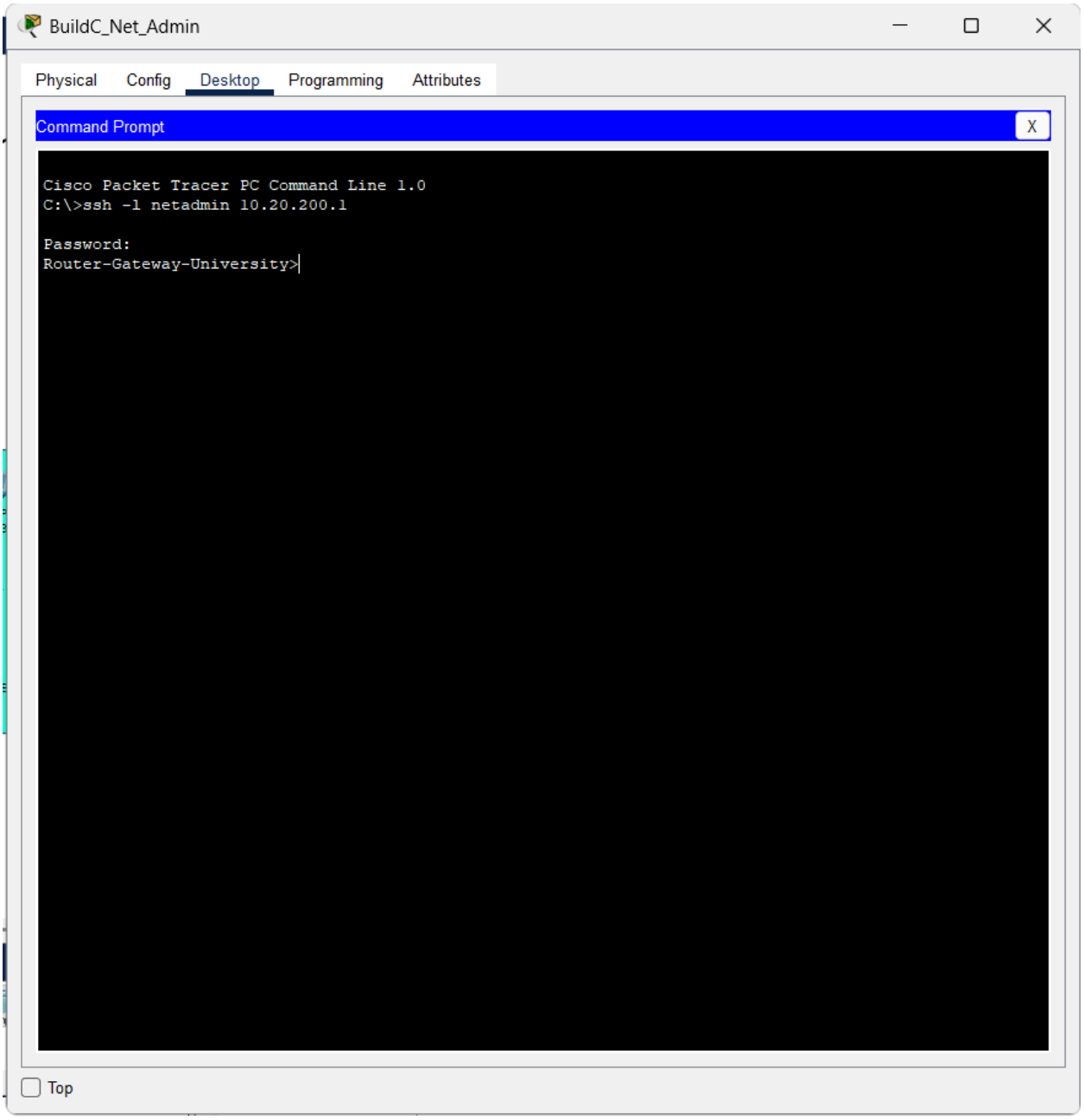


5.4. SSH Configuration (MGMT VLAN Only)

Commands to Capture

```
show ip ssh
show run | include line vty
```


Router-Gateway-University
Physical
Config
CLI
Attributes
IOS Command Line Interface
ip address 10.20.90.1 255.255.255.0
ip helper-address 10.20.60.10
ip access-group GUEST_OUT in
interface GigabitEthernet0/0.200
description VLAN 200 - Management
encapsulation dot1Q 200
ip address 10.20.200.1 255.255.255.0
ip helper-address 10.20.60.10
ip access-group MGMT_PROTECT in
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#enable
Router-Gateway-University#show run | include helper
ip helper-address 10.20.60.10
ip helper-address 10.20.60.10
ip helper-address 10.20.60.10
ip helper-address 10.20.60.10
ip helper-address 10.20.60.10
ip helper-address 10.20.60.10
ip helper-address 10.20.60.10
ip helper-address 10.20.60.10
ip helper-address 10.20.60.10
ip helper-address 10.20.60.10
ip helper-address 10.20.60.10
Router-Gateway-University#
Router-Gateway-University#
Router-Gateway-University#show ip ssh
SSH Enabled - version 1.99
Authentication timeout: 120 secs; Authentication retries: 3
Router-Gateway-University#show run | include line vty
line vty 0 4
Router-Gateway-University#
Copy
Paste
Top



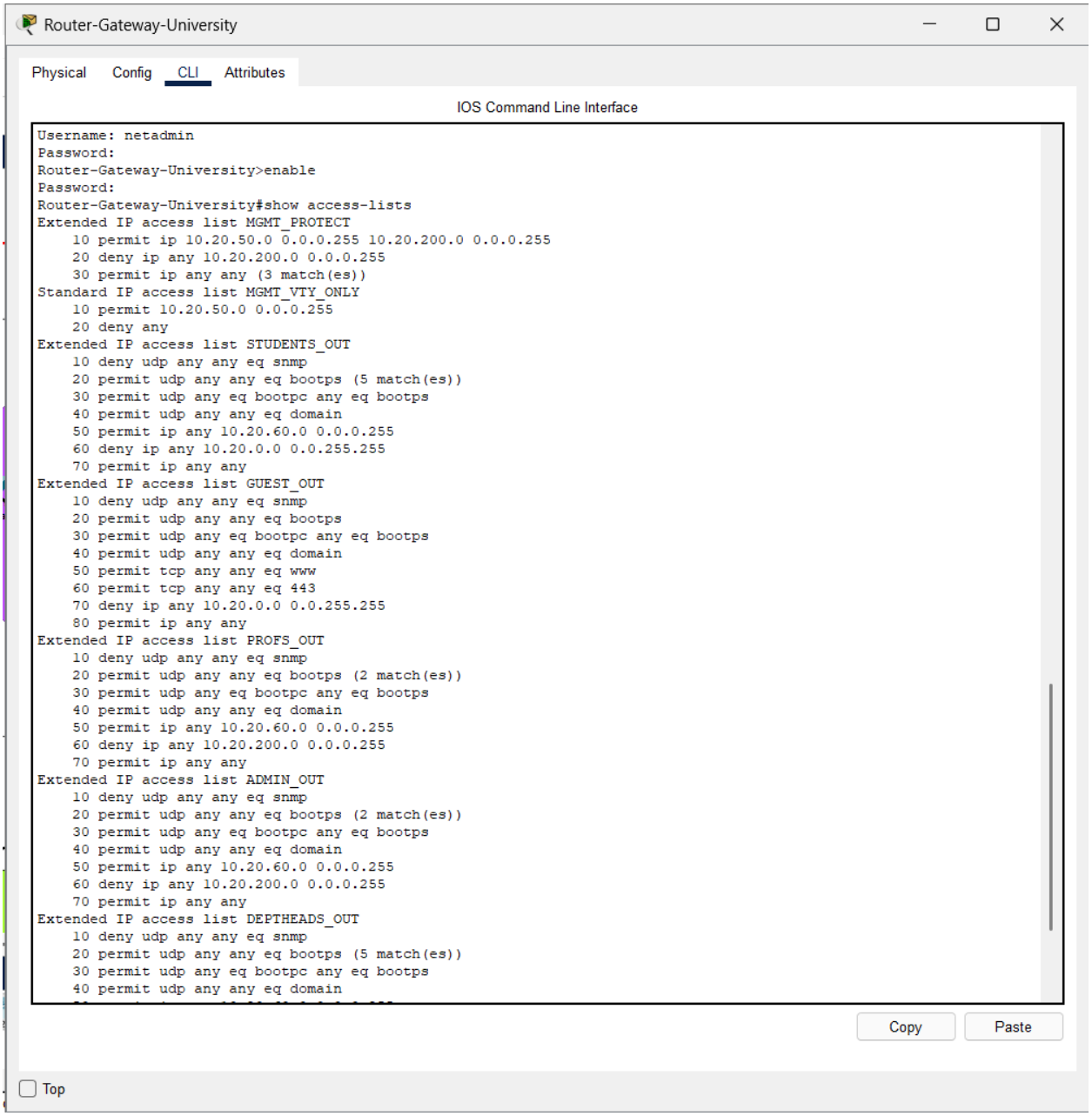
5.5. ACL Configuration & Proof

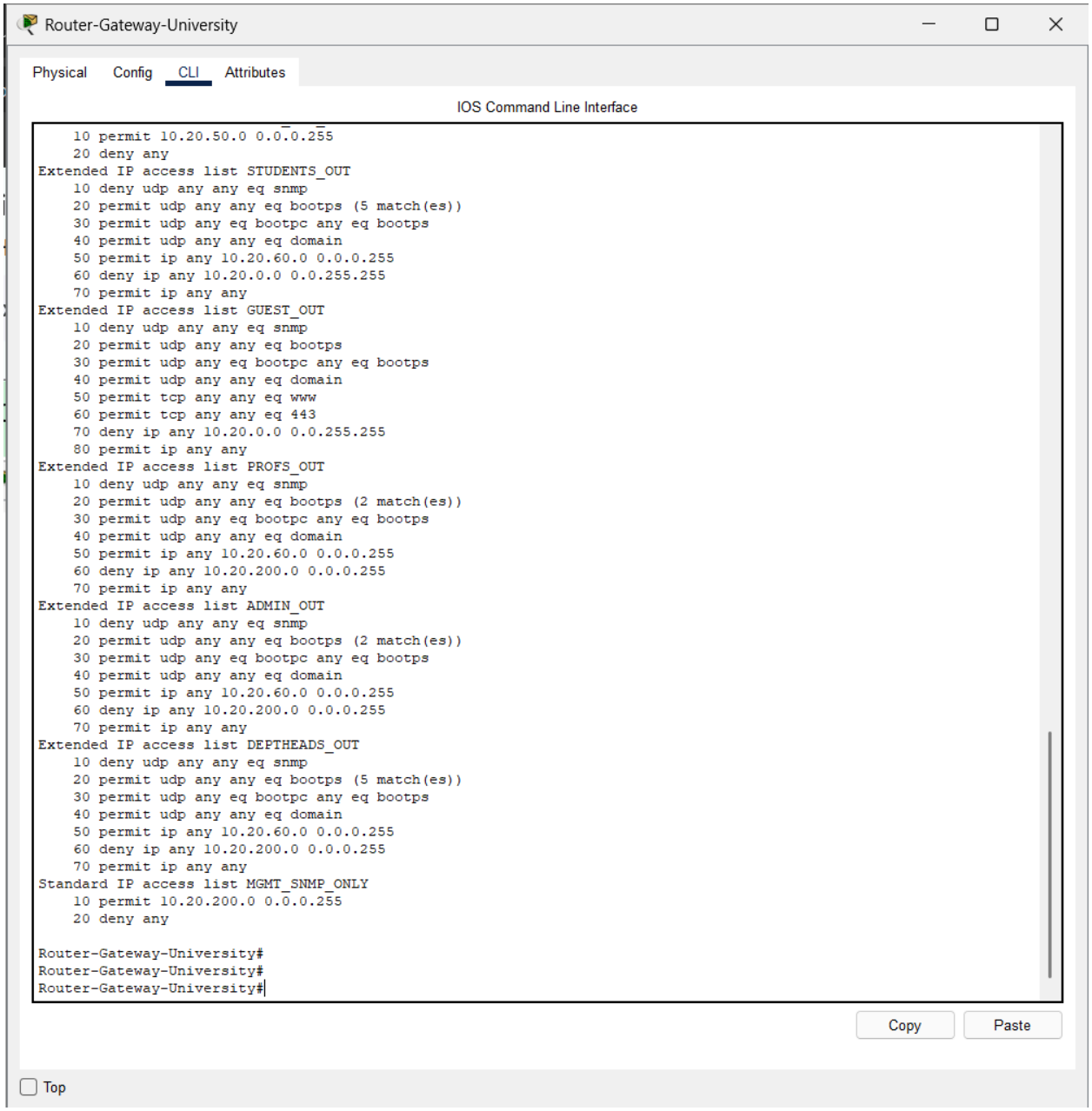
You MUST show both:

1. ACL itself
2. ACL applied to an interface
3. A traffic test proving it works

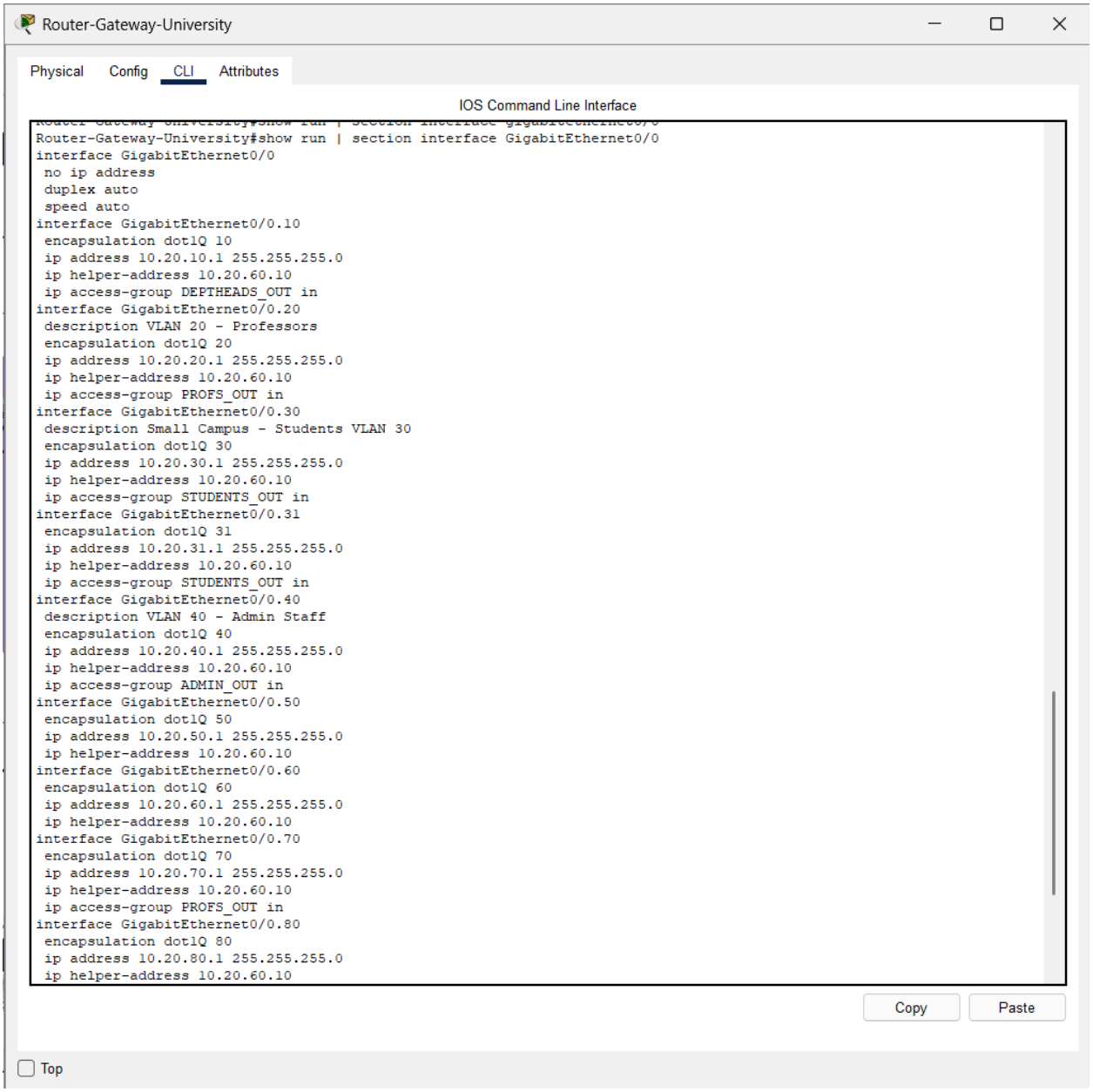
Commands to Capture

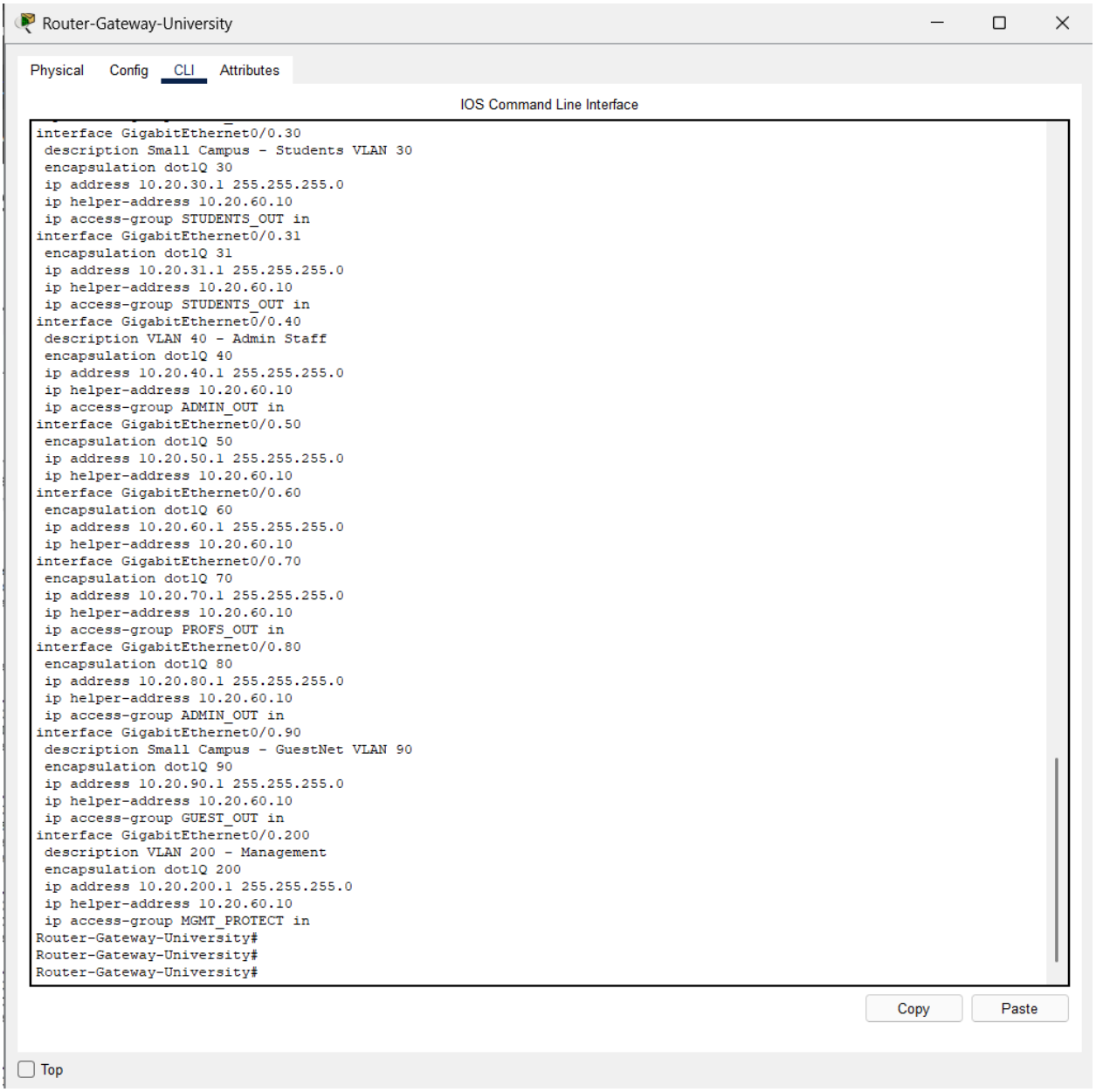
ACL definition:





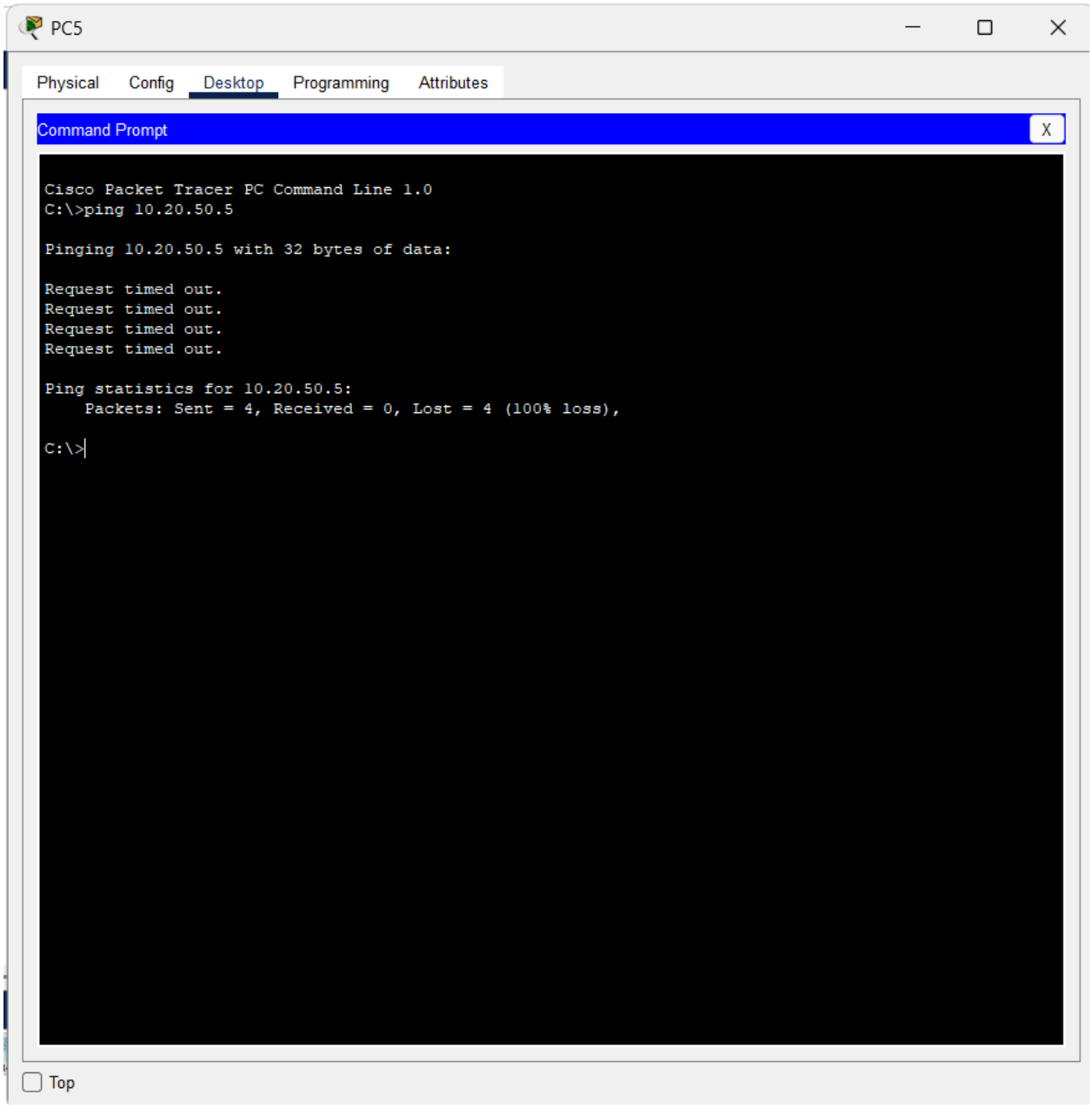
ACL applied to interface:


Router-Gateway-University
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router-Gateway-University#show run | section interface GigabitEthernet0/0
interface GigabitEthernet0/0
 no ip address
 duplex auto
 speed auto
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 10.20.10.1 255.255.255.0
 ip helper-address 10.20.60.10
 ip access-group DEPTHEADS_OUT in
interface GigabitEthernet0/0.20
 description VLAN 20 - Professors
 encapsulation dot1Q 20
 ip address 10.20.20.1 255.255.255.0
 ip helper-address 10.20.60.10
 ip access-group PROFS_OUT in
interface GigabitEthernet0/0.30
 description Small Campus - Students VLAN 30
 encapsulation dot1Q 30
 ip address 10.20.30.1 255.255.255.0
 ip helper-address 10.20.60.10
 ip access-group STUDENTS_OUT in
interface GigabitEthernet0/0.31
 encapsulation dot1Q 31
 ip address 10.20.31.1 255.255.255.0
 ip helper-address 10.20.60.10
 ip access-group STUDENTS_OUT in
interface GigabitEthernet0/0.40
 description VLAN 40 - Admin Staff
 encapsulation dot1Q 40
 ip address 10.20.40.1 255.255.255.0
 ip helper-address 10.20.60.10
 ip access-group ADMIN_OUT in
interface GigabitEthernet0/0.50
 encapsulation dot1Q 50
 ip address 10.20.50.1 255.255.255.0
 ip helper-address 10.20.60.10
interface GigabitEthernet0/0.60
 encapsulation dot1Q 60
 ip address 10.20.60.1 255.255.255.0
 ip helper-address 10.20.60.10
interface GigabitEthernet0/0.70
 encapsulation dot1Q 70
 ip address 10.20.70.1 255.255.255.0
 ip helper-address 10.20.60.10
 ip access-group PROFS_OUT in
interface GigabitEthernet0/0.80
 encapsulation dot1Q 80
 ip address 10.20.80.1 255.255.255.0
 ip helper-address 10.20.60.10
Copy
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Top

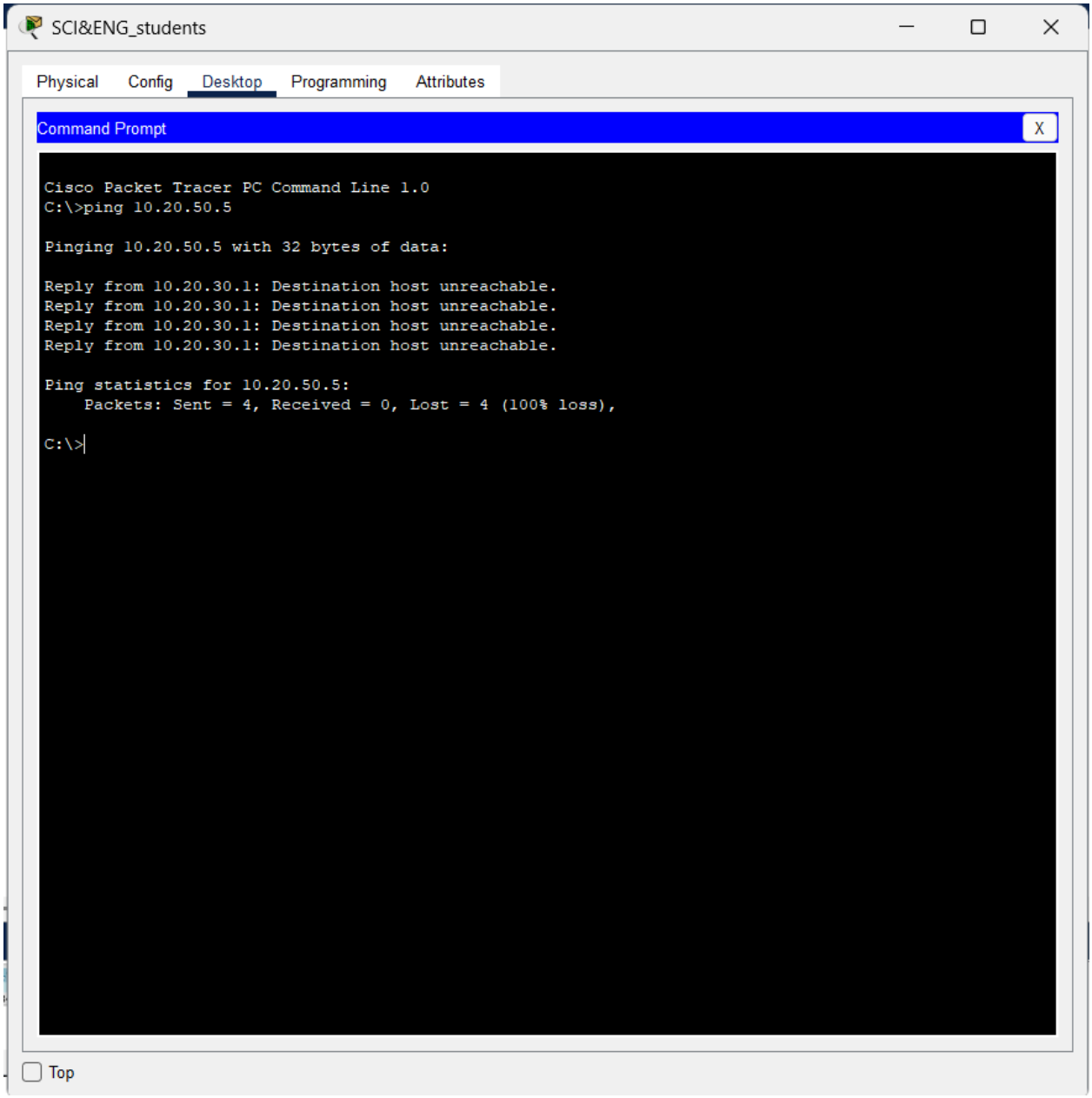


Traffic Test

From unauthorized PC:



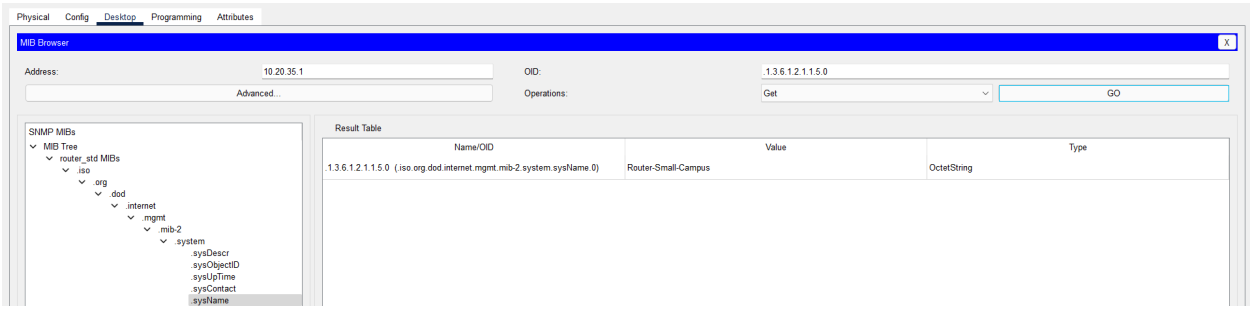
From allowed PC:



5.6. SNMP Monitoring

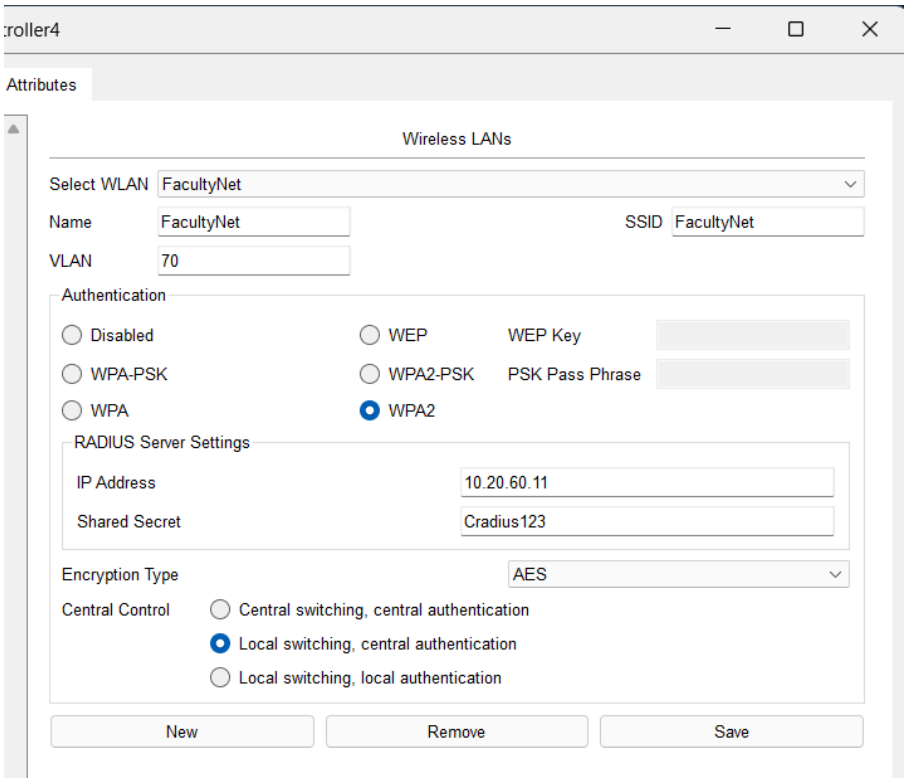
We enabled SNMP on the core routers to monitor device status and performance. A management PC on the management VLAN uses the SNMP MIB Browser to query routers (e.g.

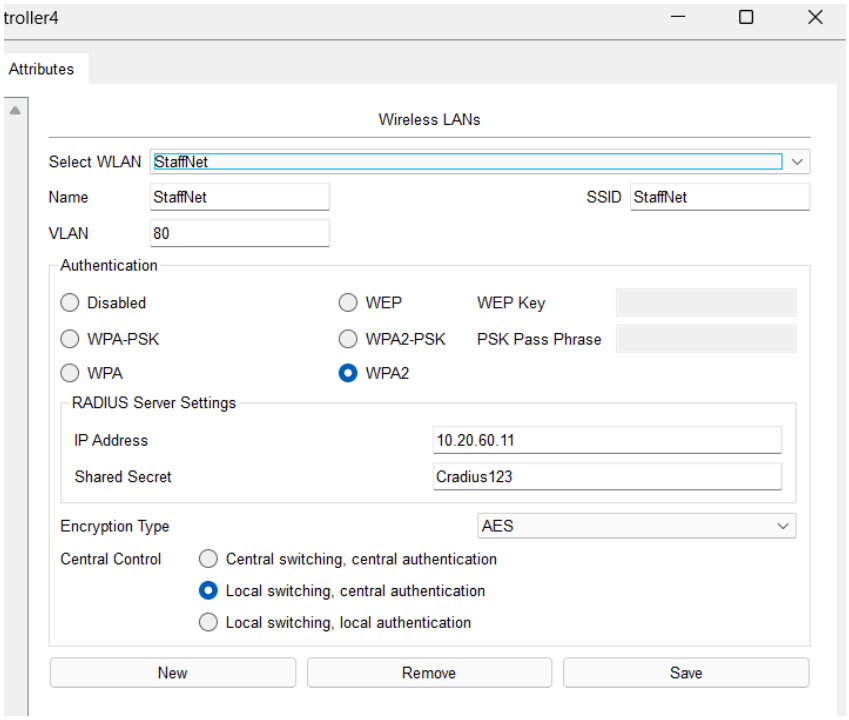
uptime, interfaces). SNMP communities are protected with ACLs so that only hosts in the management VLAN can access SNMP.



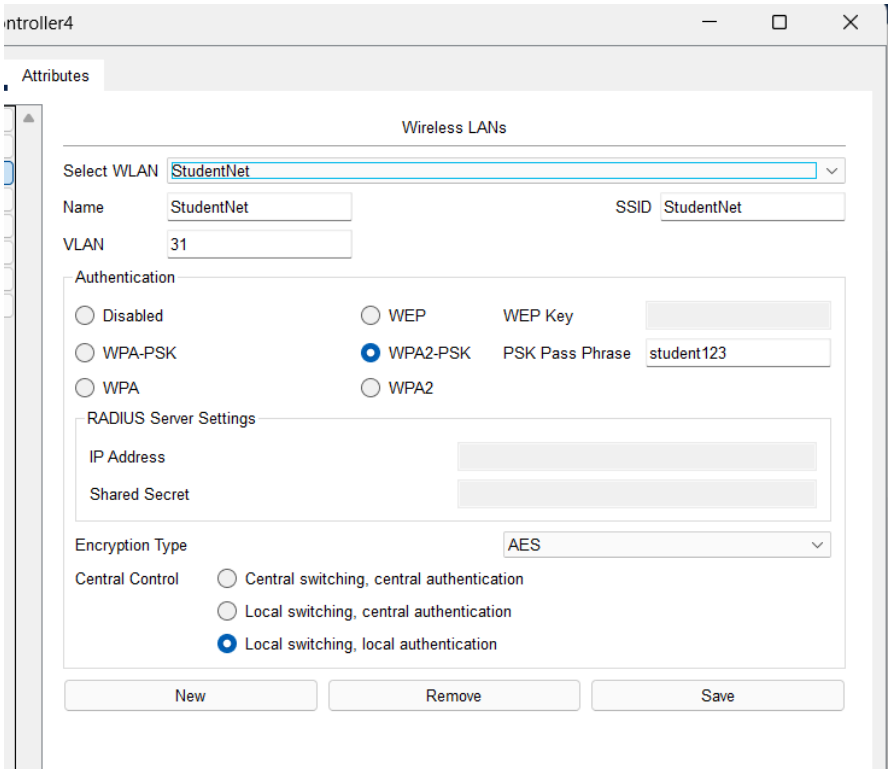
6. Wireless Configuration (WLC + APs)

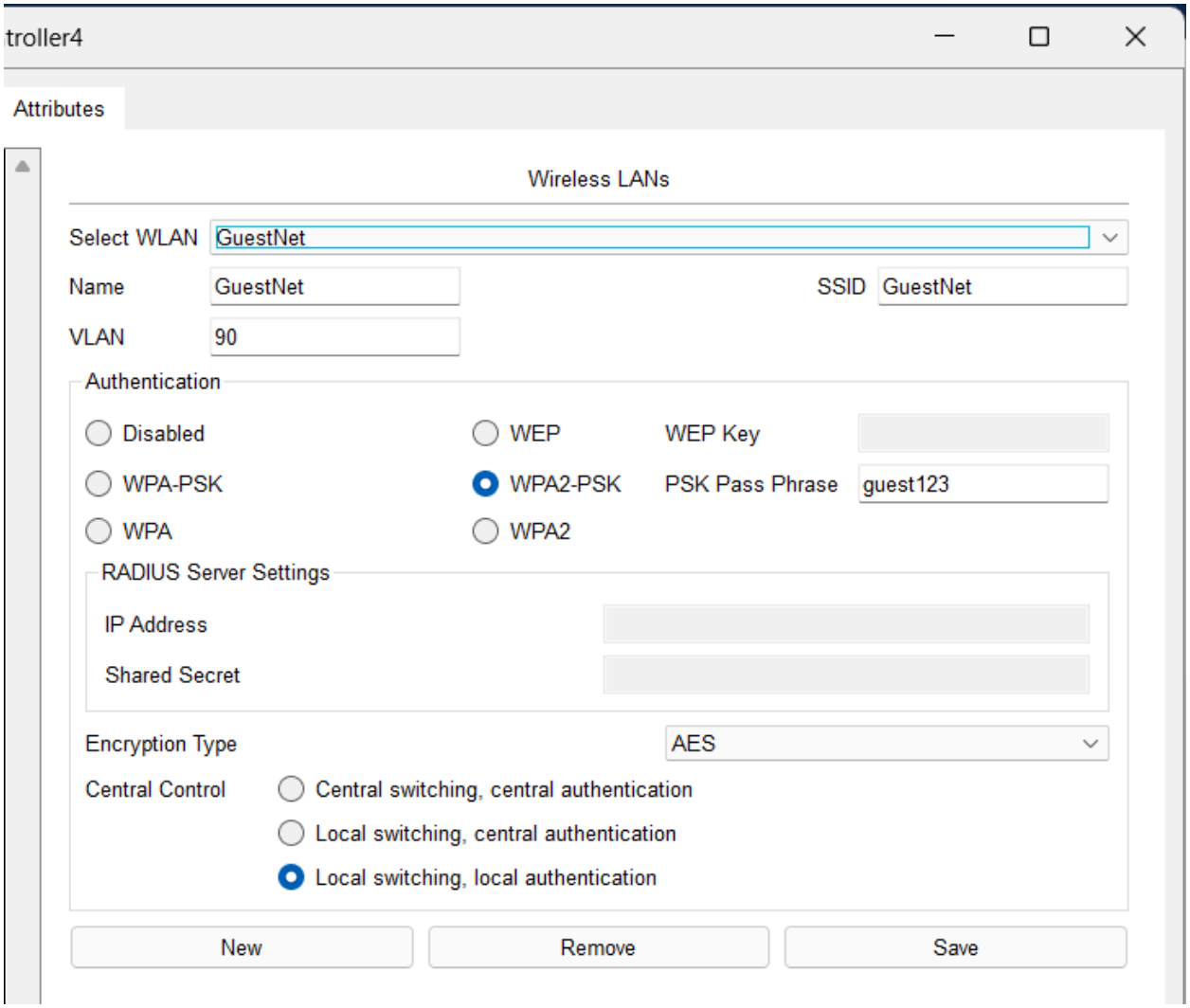
We deployed campus-wide wireless using lightweight APs controlled by WLCs in the main and small campuses. Four SSIDs are broadcast and mapped to their VLANs: **FacultyNet**, **StaffNet** (WPA2-Enterprise via RADIUS), **StudentNet** (WPA2-PSK/Enterprise), and **GuestNet** (WPA2-PSK). DHCP provides addresses per VLAN, and ACLs restrict GuestNet to internet-only access while faculty, staff, and students get full internal connectivity.



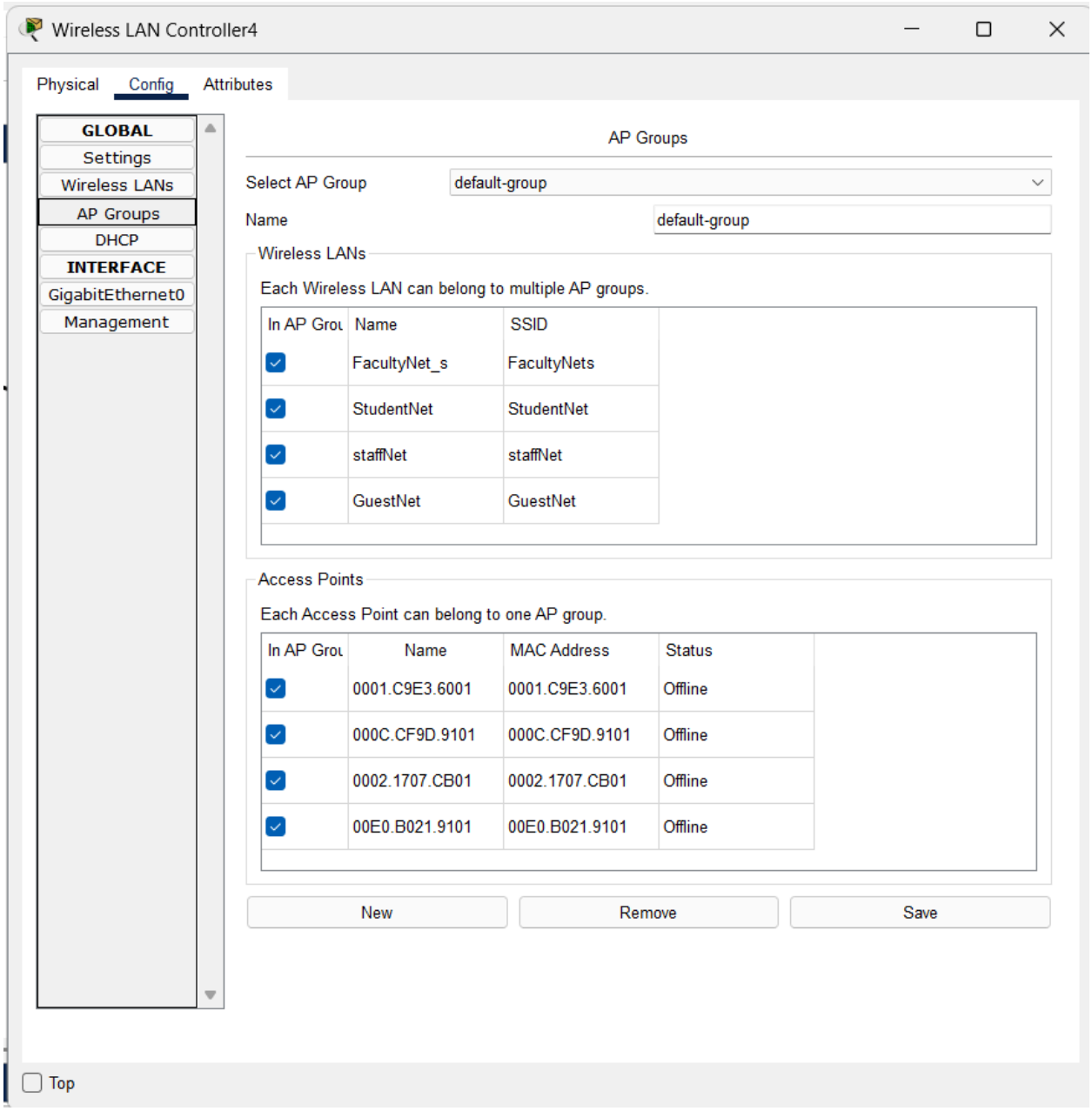


Connected with Radius server to apply WPA2-Enterprise

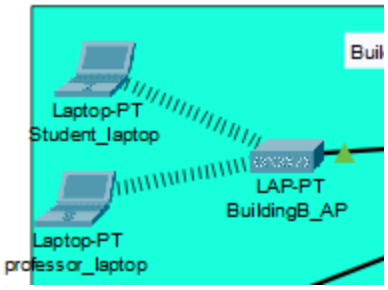



troller4
Attributes
Wireless LANs
Select WLAN
GuestNet
Name
GuestNet
SSID
GuestNet
VLAN
90
Authentication
Disabled
WEP
WEP Key
WPA-PSK
WPA2-PSK
PSK Pass Phrase
guest123
WPA
WPA2
RADIUS Server Settings
IP Address
Shared Secret
Encryption Type
AES
Central Control
Central switching, central authentication
Local switching, central authentication
Local switching, local authentication
New
Remove
Save

1. AP Join Status



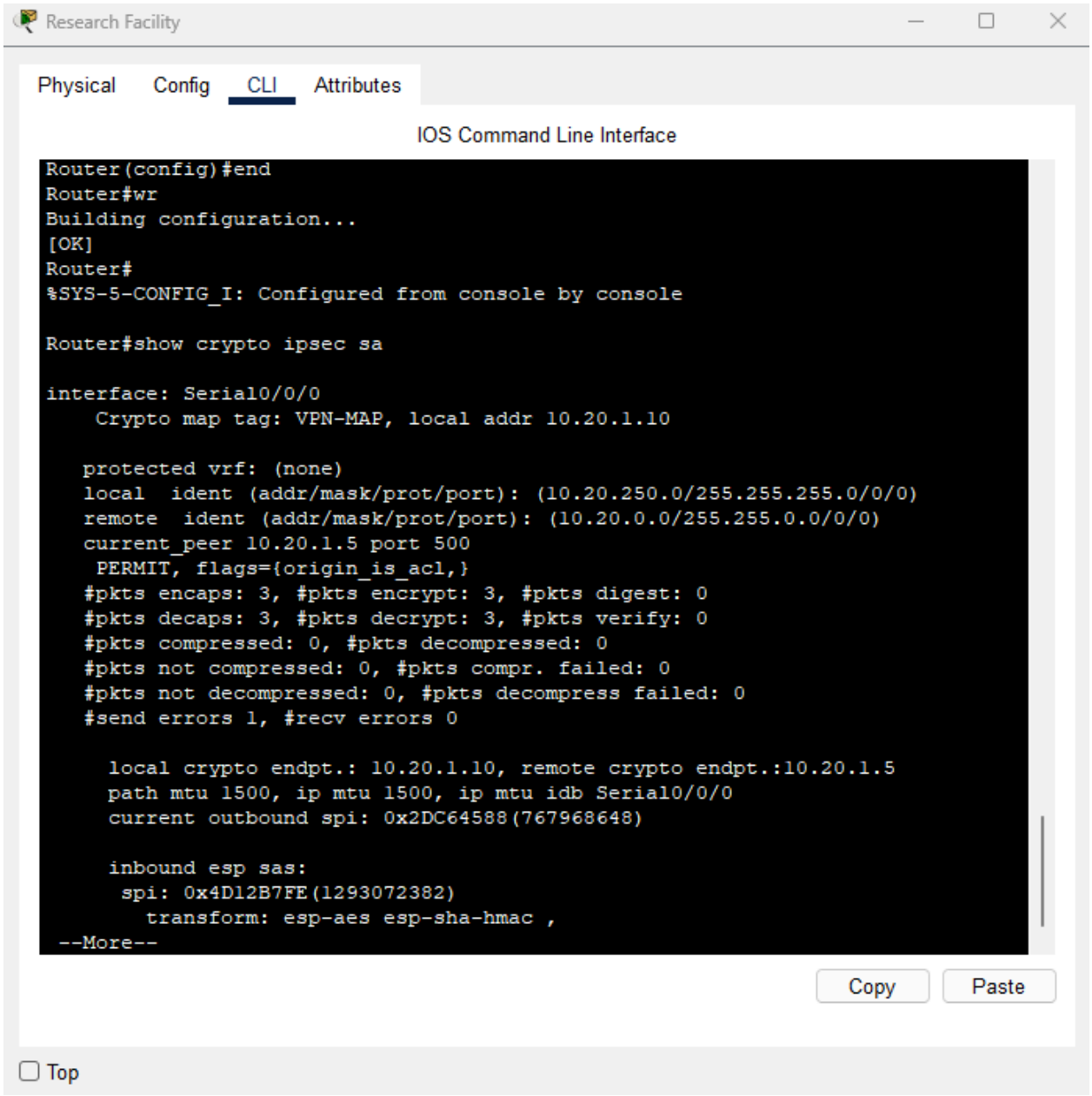
Connected Laptops



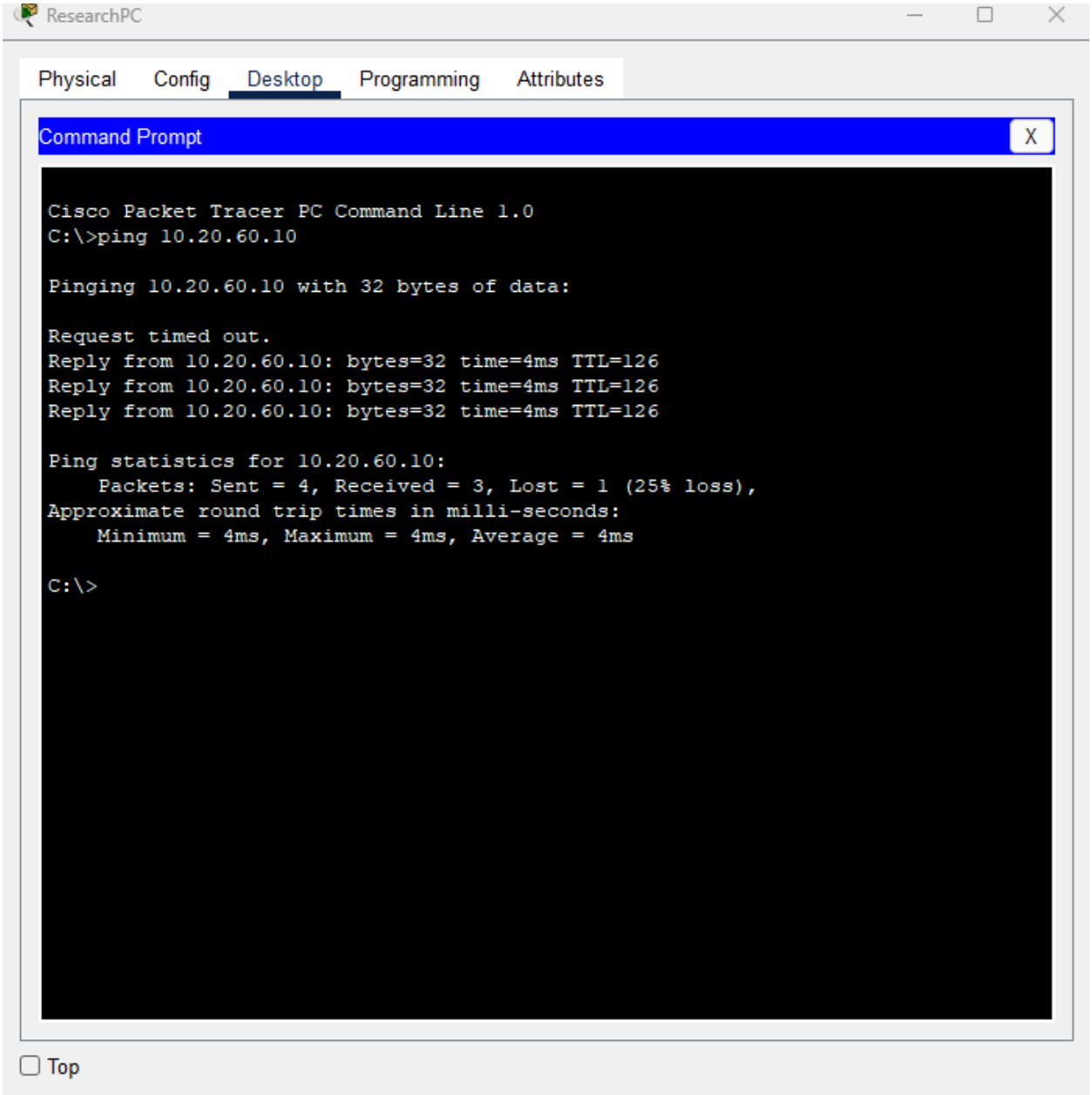
One AP can connect more than one different VLANs

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7. VPN Configuration (Research Facility ↔ Campus Gateway)



Ping Screenshot from Research PC to DHCP server in Building C



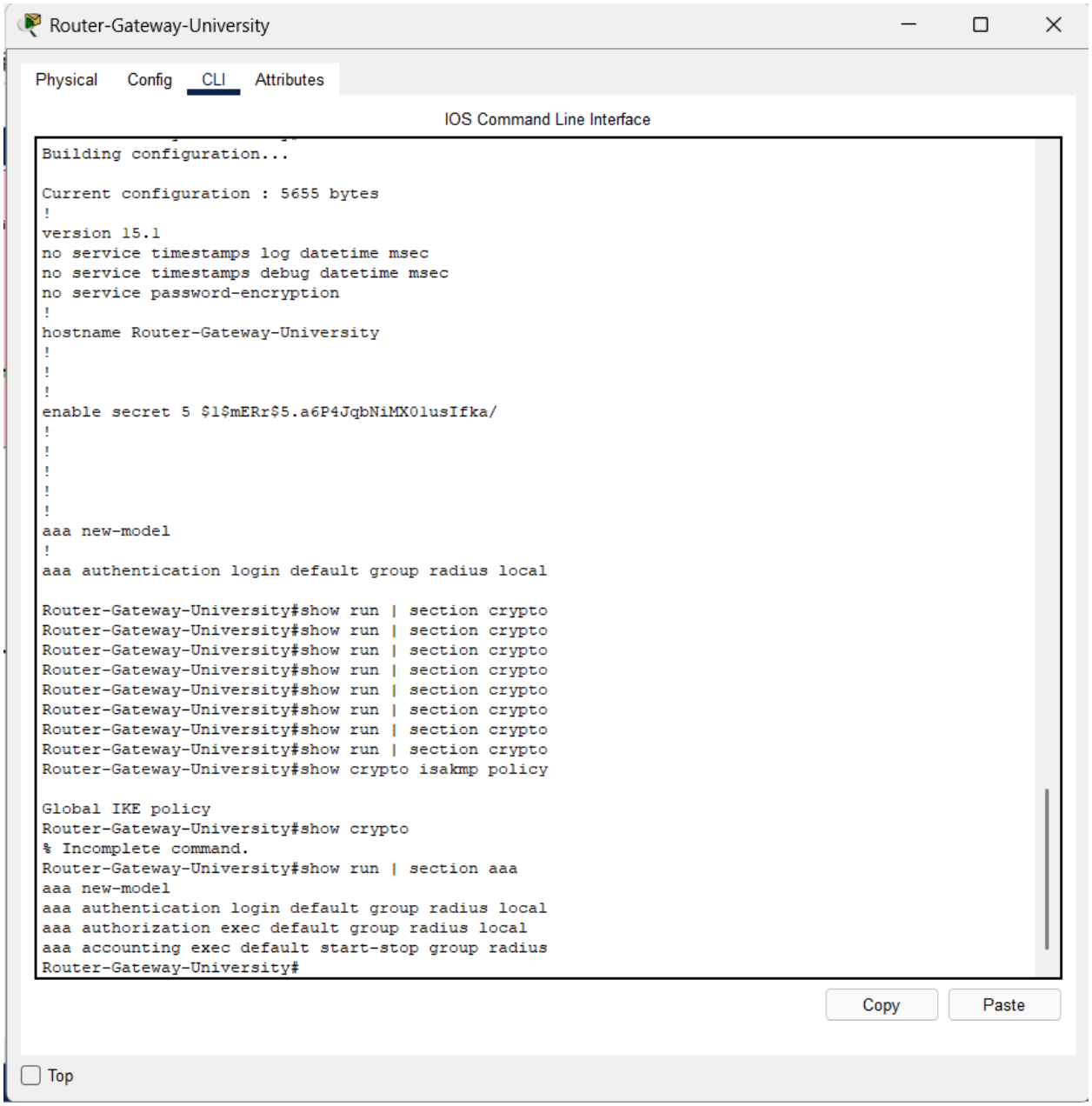
.

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8. Syslog, AAA, and Network Hardening

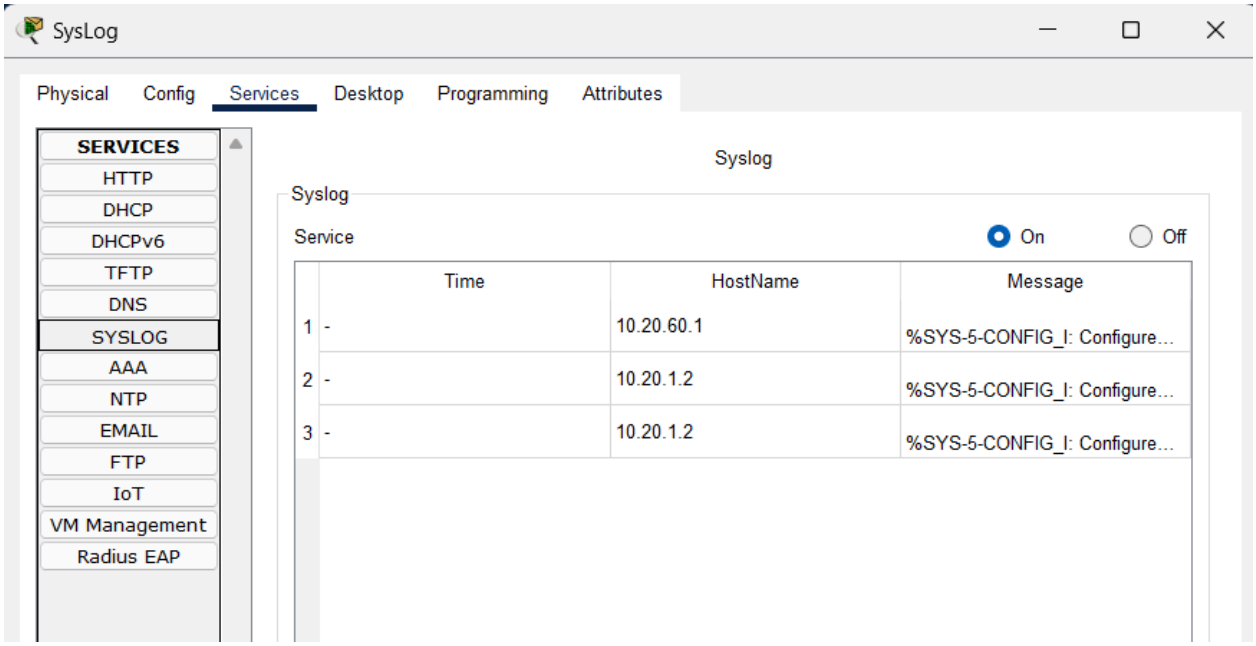
Show these configurations:

AAA on Router

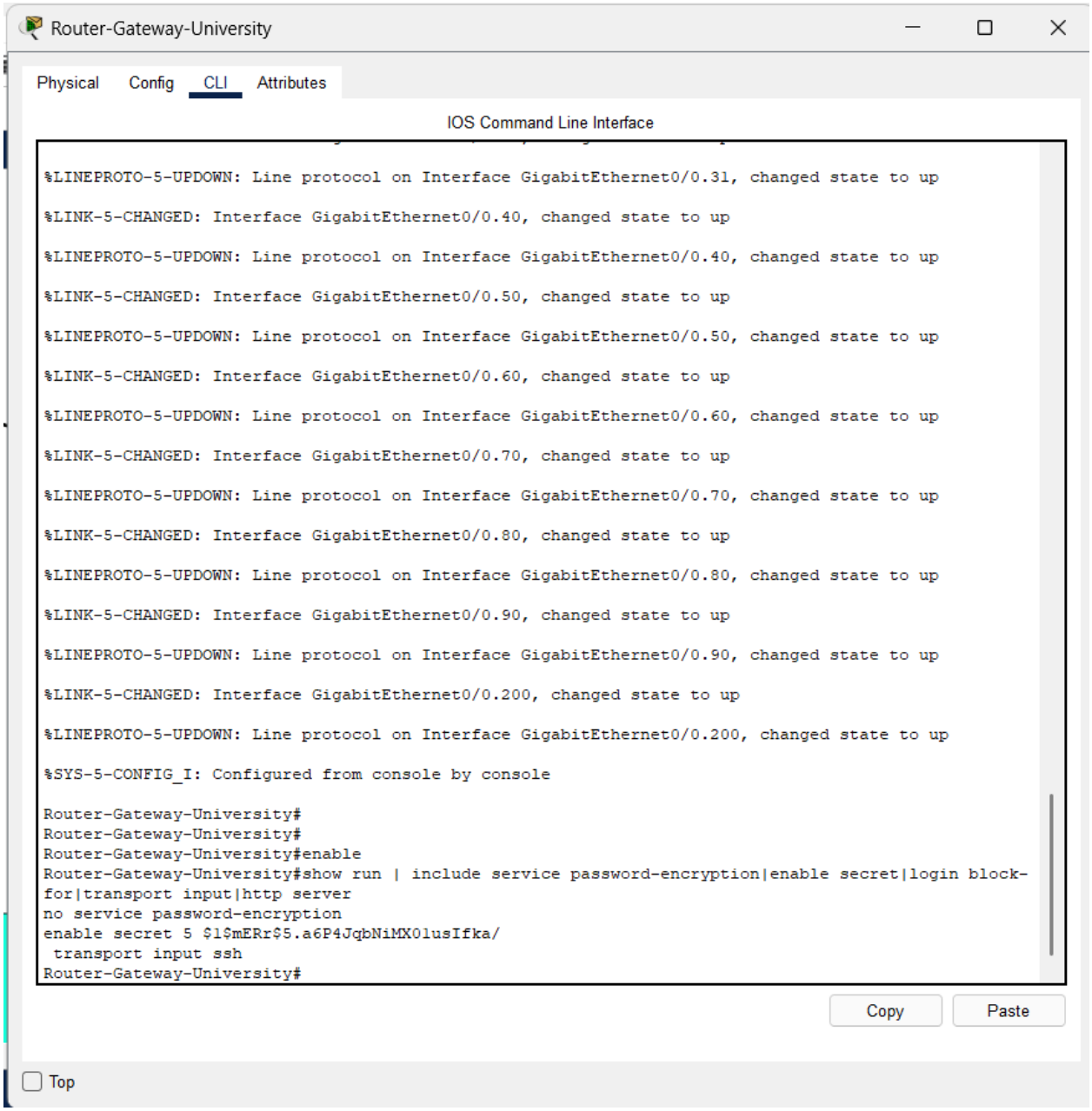


Syslog verification

We configured a central Syslog server to collect logs from all routers and key switches. Each device sends its system messages to the server using the logging command, so link changes, config edits, and security events are stored in one place. Access to the Syslog server is restricted to the management VLAN, supporting secure monitoring and easier troubleshooting.



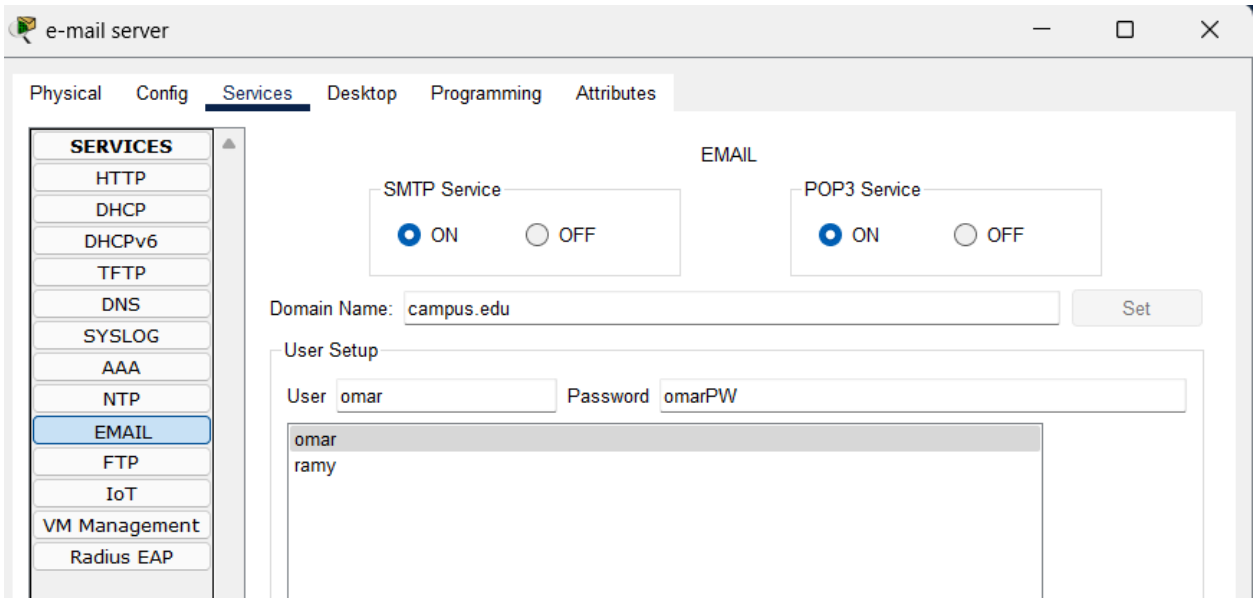
Security Hardening Commands to Show:

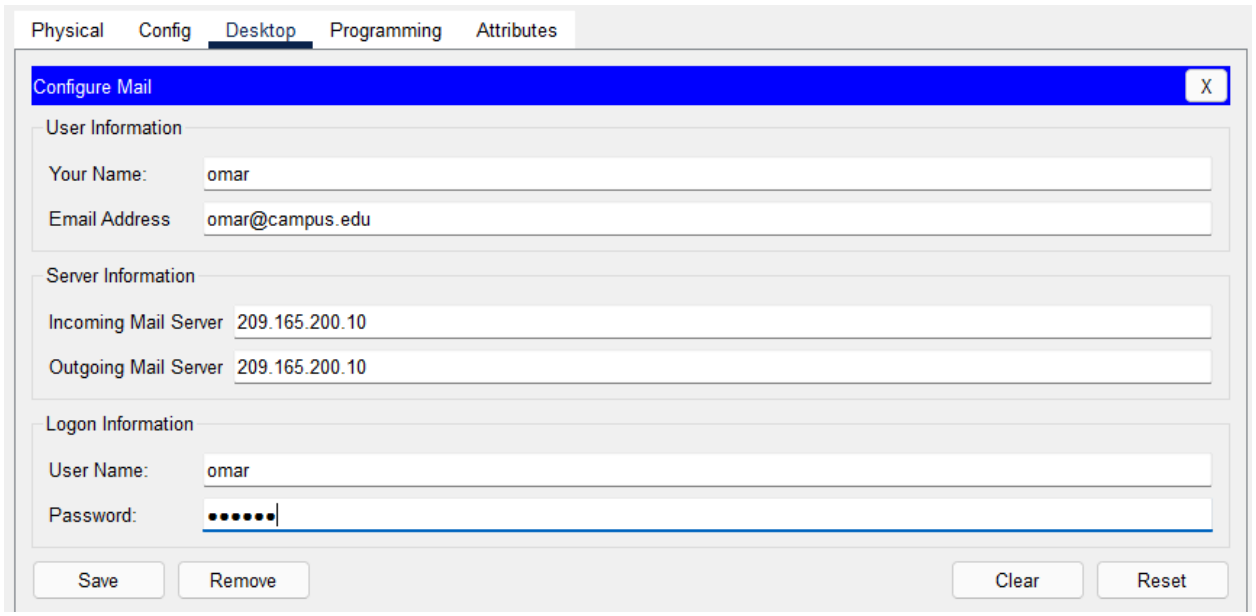


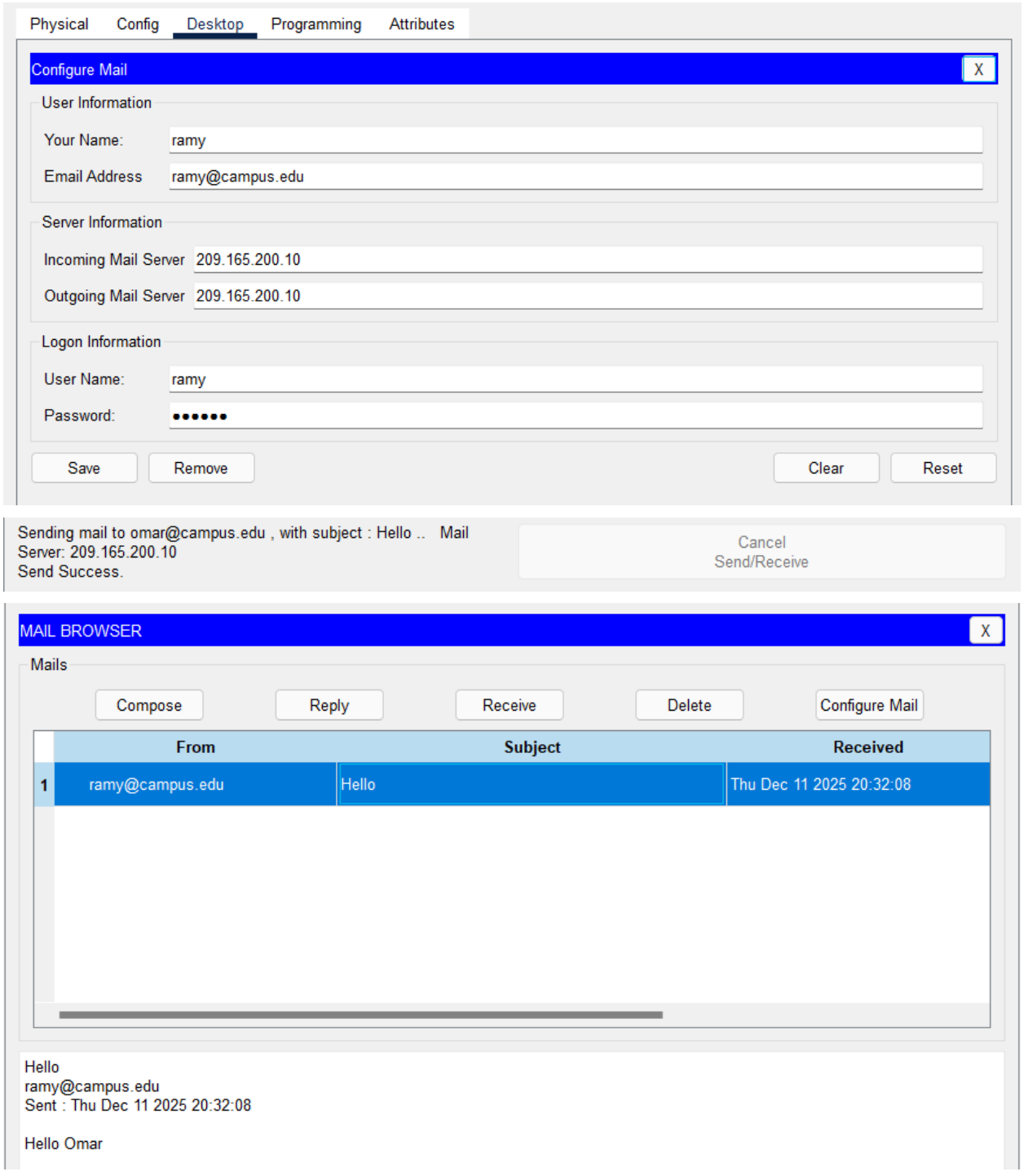
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9. Email Server

We deployed an email server using the **campus.edu** domain. SMTP and POP3 services are enabled so users can send and receive mail with standard clients. Mailboxes were created for users (e.g. *omar@campus.edu*, *ramy@campus.edu*) with individual passwords.


e-mail server
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
EMAIL
SMTP Service
ON
OFF
POP3 Service
ON
OFF
Domain Name: campus.edu
Set
User Setup
User omar
Password omarPW
omar
ramy


Physical
Config
Desktop
Programming
Attributes
Configure Mail
X
User Information
Your Name: omar
Email Address omar@campus.edu
Server Information
Incoming Mail Server 209.165.200.10
Outgoing Mail Server 209.165.200.10
Logon Information
User Name: omar
Password: ••••••
Save
Remove
Clear
Reset



---

10. Risk Assessment & Security Policies

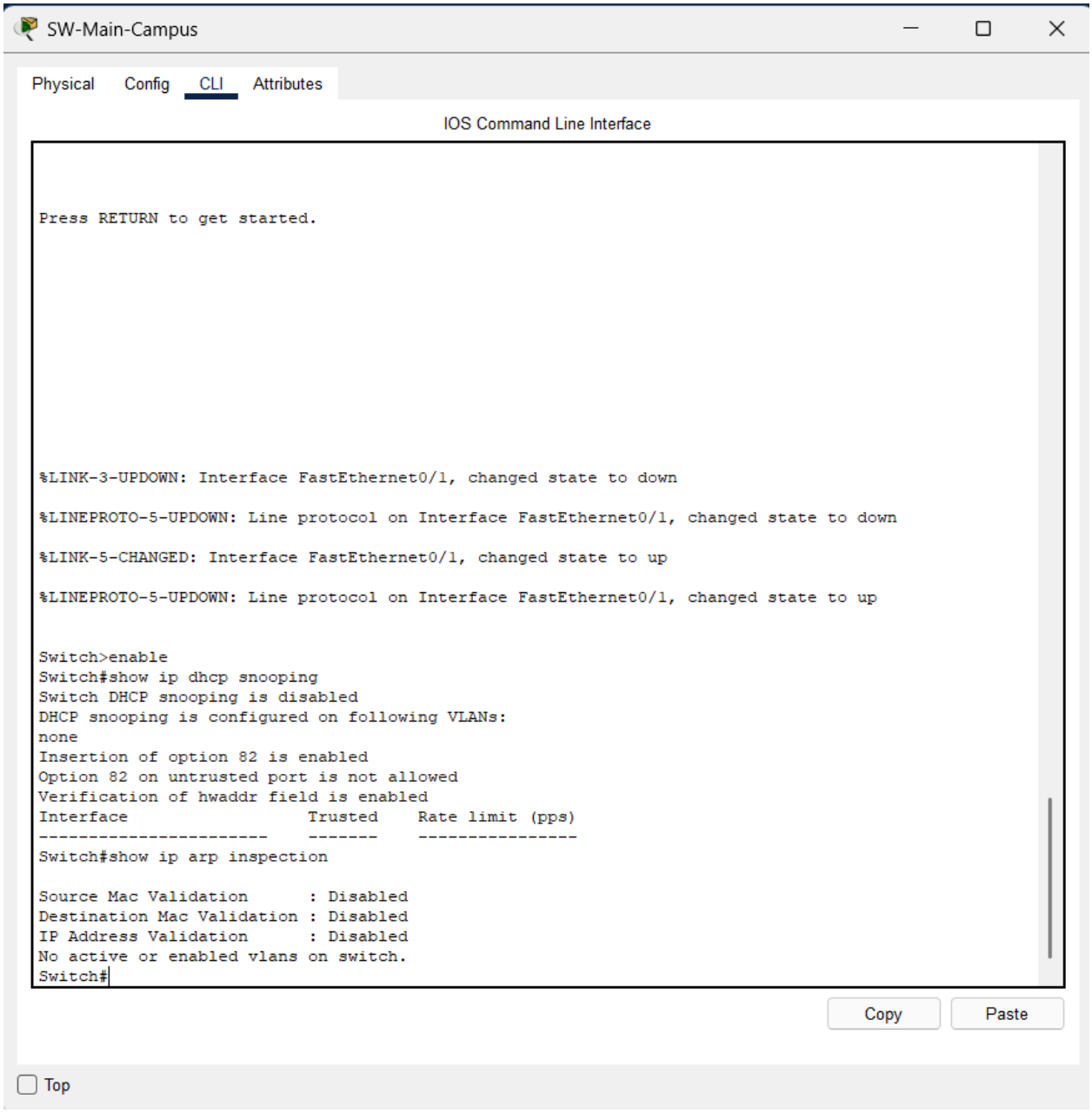
Write these in short, professional paragraphs:

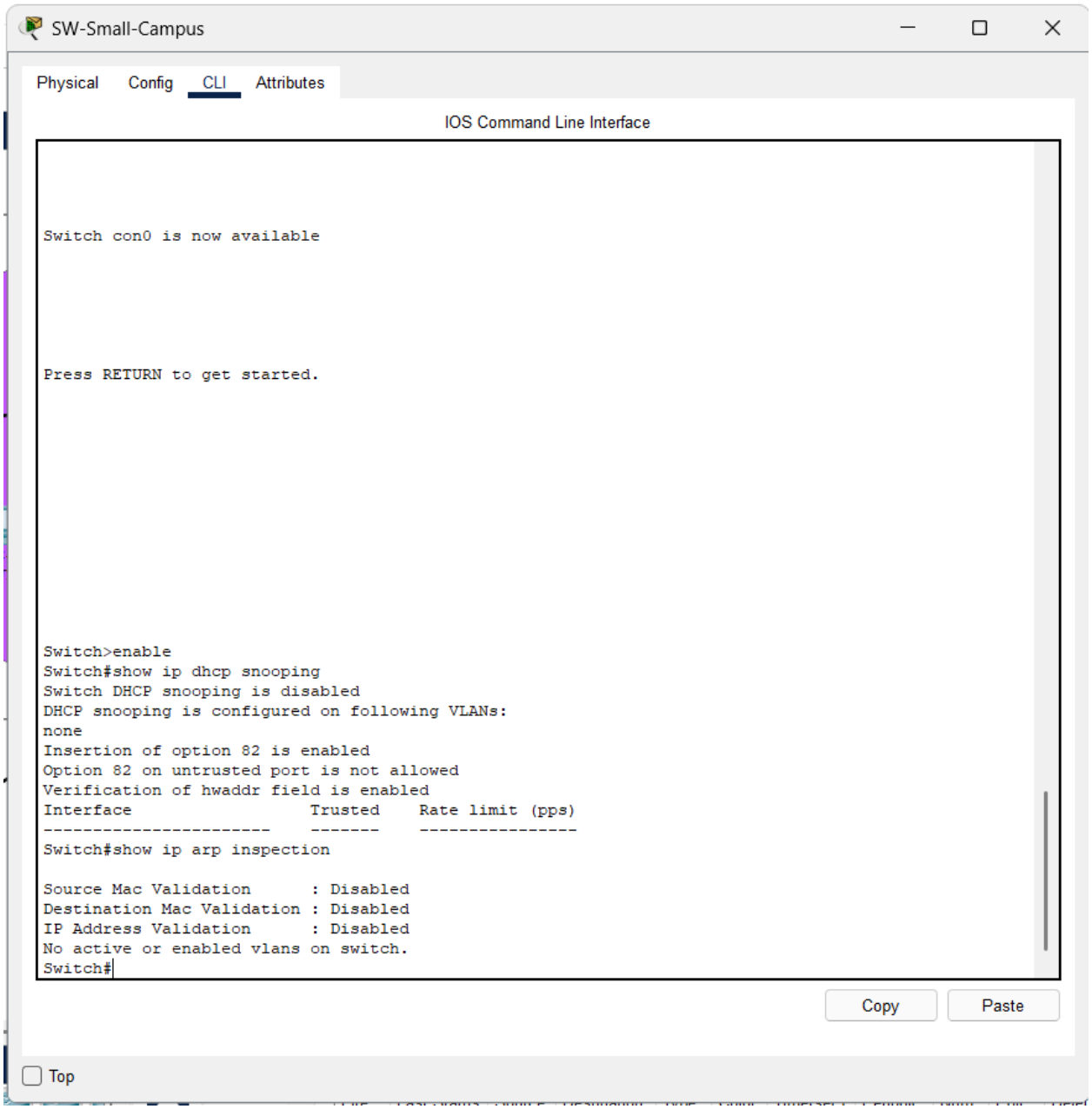
Top Risks Identified

- Unauthorized inter-VLAN access
- Rogue DHCP or ARP spoofing
- Compromised guest devices
- Weak WiFi security
- Lack of centralized logging

Policies Adopted

- ACLs between ALL user VLANs
- DHCP Snooping + DAI on all switches
- WPA2 Enterprise for staff/faculty
- Guest VLAN isolation
- SSH-only management + AAA authentication
- Syslog monitoring
- VPN encryption for remote site


SW-Main-Campus
Physical
Config
CLI
Attributes
IOS Command Line Interface
Press RETURN to get started.
%LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Switch>enable
Switch#show ip dhcp snooping
Switch DHCP snooping is disabled
DHCP snooping is configured on following VLANs:
none
Insertion of option 82 is enabled
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Interface Trusted Rate limit (pps)
Switch#show ip arp inspection
Source Mac Validation : Disabled
Destination Mac Validation : Disabled
IP Address Validation : Disabled
No active or enabled vlans on switch.
Switch#
Copy
Paste
Top



11. Conclusion

This project successfully demonstrates the design and implementation of a **secure enterprise-grade university network**, including:

- Proper VLAN segmentation

- Secure inter-VLAN routing
- Wireless infrastructure using WLC
- Centralized DHCP, AAA, Syslog
- SSH-only management
- Port security & Layer 2 protection
- Working IPSec VPN
- Simulation-mode proofs

The design meets all functional and security requirements and follows industry best practices.